

Review

A Review of the Efficient and Thermal Utilization of Biomass Waste

Jiaao Zhu ¹, Yun Guo ^{1,*} , Na Chen ² and Baoming Chen ²

¹ School of Mechanical and Automotive Engineering, Shanghai University of Engineering Science, Shanghai 201620, China

² School of Energy and Power Engineering, University of Shanghai for Science and Technology, Shanghai 200093, China

* Correspondence: graceguo1980@126.com

Abstract: As a new type of energy that can meet the requirements of carbon neutrality, biomass has received wide attention in recent years, and its rational and efficient thermal utilization can help reduce greenhouse gas emissions and establish an energy-saving, low-carbon energy system to promote sustainable development. In this paper, the current utilization and research status of plant-based biomass waste is comprehensively summarized from four aspects, namely component properties, industrial thermal utilization means, experiments and theoretical calculations. In addition, this paper summarizes the research progress in several aspects, such as microscopic experimental studies, macroscopic pyrolysis characterization, and multiscale theoretical model construction of biomass waste. However, due to the diversity and heterogeneity of biomass, there are still some challenges to extending the laboratory research results to large-scale industrial production, for which we also provide an outlook on future technological innovations and development directions in this research area.

Keywords: biomass waste; carbon neutralization; thermal utilization; pyrolysis; liquefaction; gasification; combustion



Citation: Zhu, J.; Guo, Y.; Chen, N.; Chen, B. A Review of the Efficient and Thermal Utilization of Biomass Waste. *Sustainability* **2024**, *16*, 9506. <https://doi.org/10.3390/su16219506>

Academic Editor: Cinzia Buratti

Received: 30 September 2024

Revised: 20 October 2024

Accepted: 22 October 2024

Published: 31 October 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The production of CO₂ during the energy conversion process has resulted in several problems including global warming and climate change [1,2]. To reduce carbon emissions and mitigate the escalating trend of environmental warming, carbon-neutral renewable energy and low-carbon fuels have emerged as focal points in the field of new energy research [3–5]. Recent years have seen significant progress in the use of various clean energy sources, including solar, wind, biomass, geothermal and ocean energy. This progress demonstrates a growing commitment to the transition to sustainable energy and to reducing dependence on fossil fuels [6]. Additionally, transitioning to renewable energy sources encourages the creation of new jobs, economic growth and resilience to climate change, contributing to a more sustainable and environmentally friendly energy system [7]. The scale of wind and solar power utilization is growing rapidly and is used in many countries around the world; however, wind and solar power utilization is constrained by climatic conditions and regions and is not easy to store [8]. As society develops, the reserves of biomass waste are increasing, and biofuels derived from biomass feedstocks are carbon-neutral fuels. This is because the carbon stored in biomass comes exclusively from atmospheric CO₂, and the amount of CO₂ emitted during combustion is equal to the amount absorbed by the plant during growth, so the atmospheric CO₂ load when using biomass as a fuel is negligible [9]. Various types of biomass can be adapted to diverse climates and soil conditions. For instance, sugarcane thrives in warmer climates, while soybeans and wheat can grow in colder regions. Biomass exhibits a higher energy density compared to wind and solar power, which primarily provide electricity. Furthermore,

biomass can be stored post-harvest and is less susceptible to weather fluctuations and temporal constraints, offering a more stable energy source. This, combined with its advantages of ease of storage and transportation, positions biomass as a viable alternative to traditional fossil fuels. Consequently, biomass has the potential to supply sufficient clean energy for the sustainable development of humanity in the future [10].

Biomass can be classified according to its source into four categories: plant-based, animal-based, microbial-based, and waste-based. The composition of biomass varies significantly among these sources. For instance, plant-based biomass is primarily composed of cellulose, while animal-based biomass predominantly consists of proteins, fats, and other nutrients. Among these categories, plant-based biomass (e.g., wood and agricultural residues) and animal-based biomass (e.g., animal manure) are currently the primary feedstocks used due to their wide availability and excellent renewability. Notably, plant-based biomass, which includes wood, straw, and rice husk, is the preferred choice for biomass feedstock due to its abundant resources, high energy density, established processing technologies, and superior economic efficiency. Owing to the diverse composition and nature of biomass waste, a number of different treatment and processing methods are required [11]. For biomass sources such as wood, which has a high content of lignin and oils, direct combustion can be utilized without the need for additional conversion technologies due to its high energy value. In contrast, straw-based biomass contains significant amounts of cellulose and moisture, resulting in a lower energy value. Consequently, pre-treatment is necessary during the processing of straw to enhance thermal utilization efficiency. Comprehensive utilization of biomass waste can increase its added value, promoting the rational use of resources, reducing environmental pollution and dependence on fossil fuels, and playing a key part in developing a circular economy and eco-agriculture [1,12,13]. In recent years, some researchers have carried out a large amount of work to investigate the development of biomass thermal utilization technology in a more comprehensive manner. For example, Zhang et al. [14] have comprehensively analyzed the mechanism of biomass pyrolysis gas formation, influencing parameters and product applications. In addition, the pyrolysis product properties of lignocellulosic biomass and their influencing factors were highlighted by Kan et al. [15]. Gao et al. [16] reveals the interaction mechanisms between biomass hydrochar and coal during co-pyrolysis. Also, Tchaptada et al. [17] provide a detailed overview of the research progress in this field with an eye on the co-thermal conversion properties of coal and biomass waste. Despite recent advancements, a comprehensive review of the microscopic processes and theoretical calculations involved in biomass pyrolysis reactions remains lacking. This paper aims to summarize recent research findings in the thermal utilization of biomass waste and provides a thorough analysis of both experimental studies and theoretical calculations. It specifically emphasizes the progress made in understanding microscopic reaction mechanisms and theoretical frameworks related to biomass pyrolysis. This review is significantly important for reducing greenhouse gas emissions, developing energy-efficient and low-carbon energy systems, and promoting sustainable development.

2. Basic Characteristics of Biomass Waste

It is notable that the limitations of the research direction mentioned in this paper—biomass waste (BW), mainly agricultural and forestry waste, such as various types of crop straw, wood chips, etc., which are mainly composed of cellulose, hemicellulose and lignin, and animal fats and oils and feces—are not in the scope of discussion.

In terms of the chemical structure [5,18], cellulose is an unbranched linear polymer with β -(1bis4)-D-glucopyranose units which are connected by β -1, 4-glycosidic, intramolecular and intermolecular hydrogen bonds [19], as shown in Figure 1. The degree of polymerization of cellulose reflects to some extent the resilience of the biomass; the longer the molecular chain of cellulose, the better its stability, and conversely, the shorter the chain, the more likely that cellulose decomposition will occur [20]. Hemicellulose is an irregular organic polymer consisting of anhydrous sugar molecules [21]. It contains short

chain side chains and is composed of many diverse types of simple sugars, consisting of pentose, glucose, galactose, mannose, and many other hexose and aldose monomers [22]. Unlike cellulose and hemicellulose, lignin consists of a branched amorphous matrix of phenolic monomers connected by α -O-4, β -O-4, β -5 and β - β bonds. Butyryl, guaia-cyl and p-hydroxyphenol of lignin are the main phenolic monomers, which constitute a complex structure containing multiple groups of hydroxyl, methoxy and carbonyl functionalities [23,24].

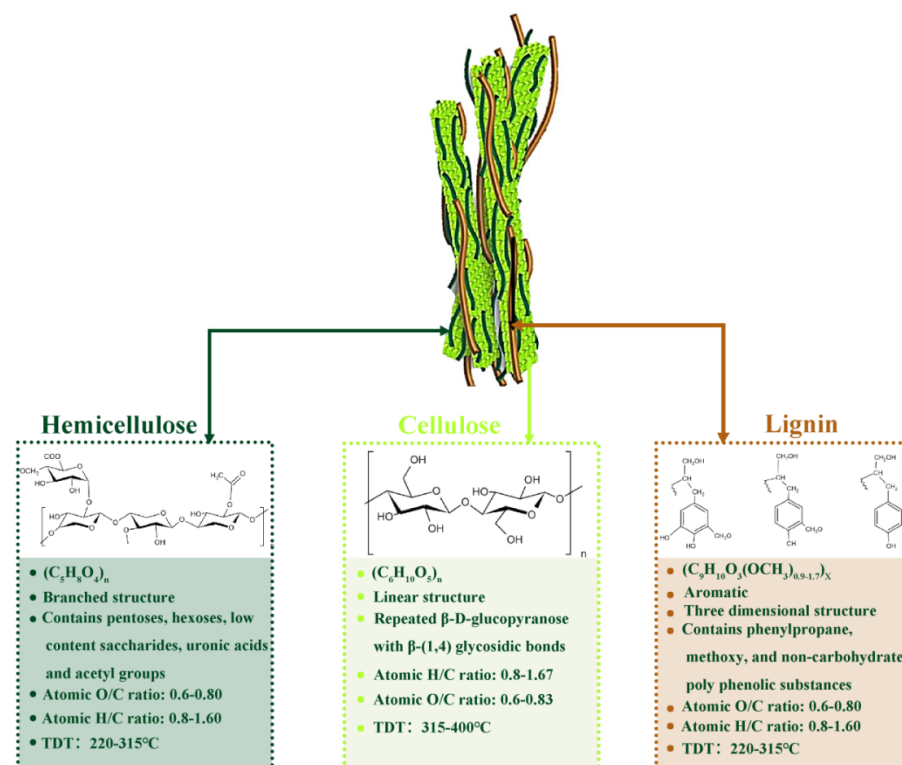


Figure 1. Typical molecular structures diagrams of hemicellulose, cellulose, and lignin.

The characteristics of some kinds of BW are shown in Table 1, which is mainly composed of lignin, cellulose and hemicellulose, and Table 2 summarizes the results of ultimate and proximate analysis.

Table 1. Characteristics of biomass waste ^a [25].

Feedstock	Cellulose	Hemicellulose	Lignin	Extractive	Ash	Ref.
Olive	34.40	36.60	21.00	7.50	0.50	[26,27]
Spruce	43.00	29.40	27.60	1.70	0.60	[28]
Beech	44.20	33.50	21.80	2.60	0.50	[29]
Cedar	37.90	22.70	33.10	3.40	0.30	[30]
Elm wood	35.92	23.75	36.34	9.15	-	[31]
Bark	38.10	23.00	32.00	4.90	2.00	[31]

^a: The composition and elemental content are influenced by various factors, such as the source of raw materials, processing techniques, and treatment methods.

Table 2. Composition and elemental analysis of biomass waste ^a.

Feedstock	Ultimate Analysis (wt%)					Proximate Analysis (wt%)			Ref.
	C	H	O	N	S	Ash	Moisture	Organic	
Fruit	50.8	6.06	42.70	42.70	0.07	1.84	19.8	78.40	[32]
Stalks	43.30	5.62	50.35	0.61	0.12	9.80	1.62	73.74	[33]

Table 2. Cont.

Feedstock	Ultimate Analysis (wt%)					Proximate Analysis (wt%)			Ref.
	C	H	O	N	S	Ash	Moisture	Organic	
Corn Stover	44.39	5.74	49.45	0.37	0.05	8.52	0.17	75.14	[34]
Plastic	75.40	11.70	0	0	12.9	0	0	100.00	[35]
Palm kernel shell	52.00	6.07	41.55	0.28	0	0.10	-	-	[36]
Sweetgum	45.18	6.59	48.23	0	0	0.45	8.78	78.14	[37]
Chlorella	60.46	9.08	27.36	3.10	0	2.20	5.70	80.80	[38]
Acacia Bark	48.90	5.97	43.90	0.82	0.15	6.37	12.70	80.90	[39]
Banana trunk	41.78	5.36	52.26	0.60	0	9.52	2.28	78.83	[40]
Palm oil sludge	50.20	6.50	34.60	6.50	0.85	17.20	43.60	39.20	[41]
Food wastes	42.00	5.40	39.30	0.40	0	12.90	1.10	70.50	[42]

^a: The composition and elemental content are influenced by various factors, such as the source of raw materials, processing techniques, and treatment methods.

3. Thermal Utilization Methods

Currently, the utilization of BW mainly contains physical treatment and biochemical and thermochemical conversion, as shown in Figure 2. However, the physical one has a relatively long processing time, and the chemical treatment method has a greater negative impact on the surrounding environment, for example, the production of irritating odor gases and sewage can affect the quality of life of the surrounding residents [43]. Biomass thermochemical conversion refers to the process of converting the chemical energy of biomass into other forms of energy by means of heat, such as pyrolysis and combustion, which has a high energy conversion efficiency and the ability to adjust the reaction conditions to produce a wide range of different products [44]. In comparison to other biomass utilization methods, thermochemical conversion offers advantages such as high efficiency, flexibility, and effective resource utilization. As a result, it is currently the predominant approach for biomass utilization. Within the realm of thermochemical conversion, various pyrolysis methods exhibit distinct advantages and applicability for biomass treatment. The four primary types of pyrolysis—namely, pyrolysis, liquefaction, gasification, and combustion—generate different main products. Consequently, it is essential to select the appropriate pyrolysis method based on specific practical applications. This section provides an overview of these four different methods of thermochemical utilization, detailing their underlying principles, processes and performance in practical applications.

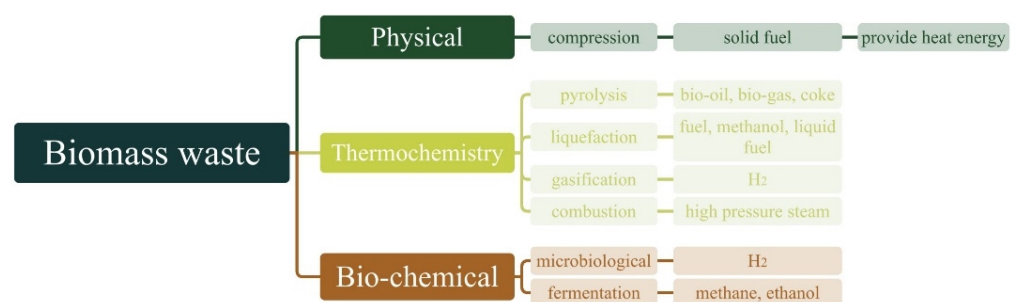


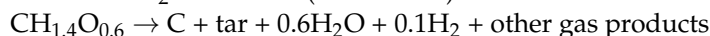
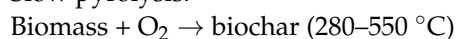
Figure 2. Main utilization methods of biomass waste.

3.1. Pyrolysis

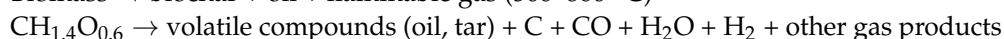
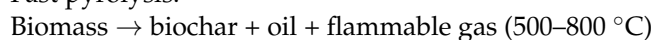
Pyrolysis is a process in which biomass feedstock is heated in an anoxic or low-oxygen environment to break down its chemical components to produce solid, liquid, and gaseous products. Pyrolysis is usually carried out in the range of 200 °C to 800 °C, depending on the physicochemical properties of the particular biomass [4,45]. The pyrolysis process can be divided into three distinct phases: the heating phase, the decomposition phase, and the cooling phase. During the heating phase, the biomass feedstock is gradually heated to a predetermined temperature, facilitating the disruption of its internal chemical bonds. In the

subsequent decomposition phase, the biomass undergoes thermal decomposition, resulting in the generation of volatile and non-volatile components at elevated temperatures. Upon completion of the reaction, the volatile products are collected in the form of gaseous or liquid outputs through cooling, while solid residues are retained. Furthermore, biomass pyrolysis can be classified into various subcategories based on differing operating conditions. Typically, pyrolysis is broadly categorized into slow pyrolysis and fast pyrolysis, primarily distinguished by the rate of temperature increase [46], corresponding to the following reaction equations [47]:

Slow pyrolysis:



Fast pyrolysis:



As can be seen from Table 3, slow pyrolysis is usually carried out at relatively low temperatures (<500 °C) and long durations (15–60 min) and is mainly used to produce biochar, which is well suited for the treatment of some high-temperature sensitive BW [48–52]. Fast pyrolysis has a high heating rate (>500 °C/s) and a short reactant residence time (<2 s), and usually produces more gases and liquids and relatively low biochar yields, which is suitable to produce both liquid and gaseous fuels (where about 50–75% of liquid fuels are produced) [53–55]. Intermediate pyrolysis, with reaction temperatures between 300–450 °C, combines the advantages of fast and slow pyrolysis. In addition, mild pyrolysis offers unique advantages in cases where the treatment of wastes requires special conditions (e.g., retention of specific chemical constituents) [56,57].

Table 3. Classification of pyrolysis technologies based on process conditions [5].

Type	Operating Conditions			Main Products	Ref.
	Temperature	Residence Time	Heat Rate		
Torrefaction (mild pyrolysis)	200–300 °C	25–35 h	>50 °C/min	Solid product, condensable liquid and non-condensable gases	[58–60]
Slow pyrolysis	280–550 °C	15–60 min	5–10 °C/min (very low)	Bio-oil (20–50%) Bio-char (20–35%) Gas (20–50%)	[61,62]
Intermediate pyrolysis	300–450 °C	About 10 min	>200–300 °C/min (Medium)	Bio-oil (35–50%) Bio-char (25–40%) Gas (20–30%)	[63]
Fast pyrolysis	500–800 °C	<2 s	>500 °C/s	Bio-oil (50–75%) Bio-char (10–25%) Gas (10–30%)	[64]

The pyrolysis pathways of cellulose, hemicellulose and lignin within different temperature ranges are shown in Figure 3. Cellulose decomposes at around 300 °C forming low molecular compounds including formic acid, furan, acetaldehyde, hydroxyacetaldehyde, H₂O and CO₂ [65]. Further degradation and depolymerization occurs at elevated temperatures up to 400–650 °C forming anhydrous sugars (mainly levoglucan) as well as other dehydrated units [66]. Levoglucan undergoes reorganization and polymerization reactions to form reactive functional groups such as -COOH, -COO-, -C=O, -OH and C-O-C. The pyrolysis temperature of hemicellulose ranges from 220–315 °C, and low molecular compounds such as acetaldehyde, furan, hydroxyacetaldehyde, formic acid, H₂O and CO₂ are formed through decomposition and polymerization reactions [65]. With a further increase in pyrolysis temperature, xylose decomposes into formic acid, acetic acid, furfural, furan, and other light oxygenated compounds. Lignin is a polymer consisting mainly of

propylphenol units with hydroxyl and methoxy substituents [67]. During pyrolysis, the methoxy group in lignin further decomposes and generates small molecule radicals, which stabilize the large molecule fragments and prevent them from polymerizing into char [68]. Li et al. [69] carbonized lignin in a high-temperature furnace after heat stabilization in a muffle furnace, and TGA analysis showed that there were three major peaks in the pyrolysis of lignin. The first one is the primary pyrolysis reaction at 350 °C, then the methoxy-related reaction from 400 °C to 450 °C, and the last peak appears between 550 °C and 600 °C and is associated with the gasification of catechol (1,2-dihydroxybenzene).

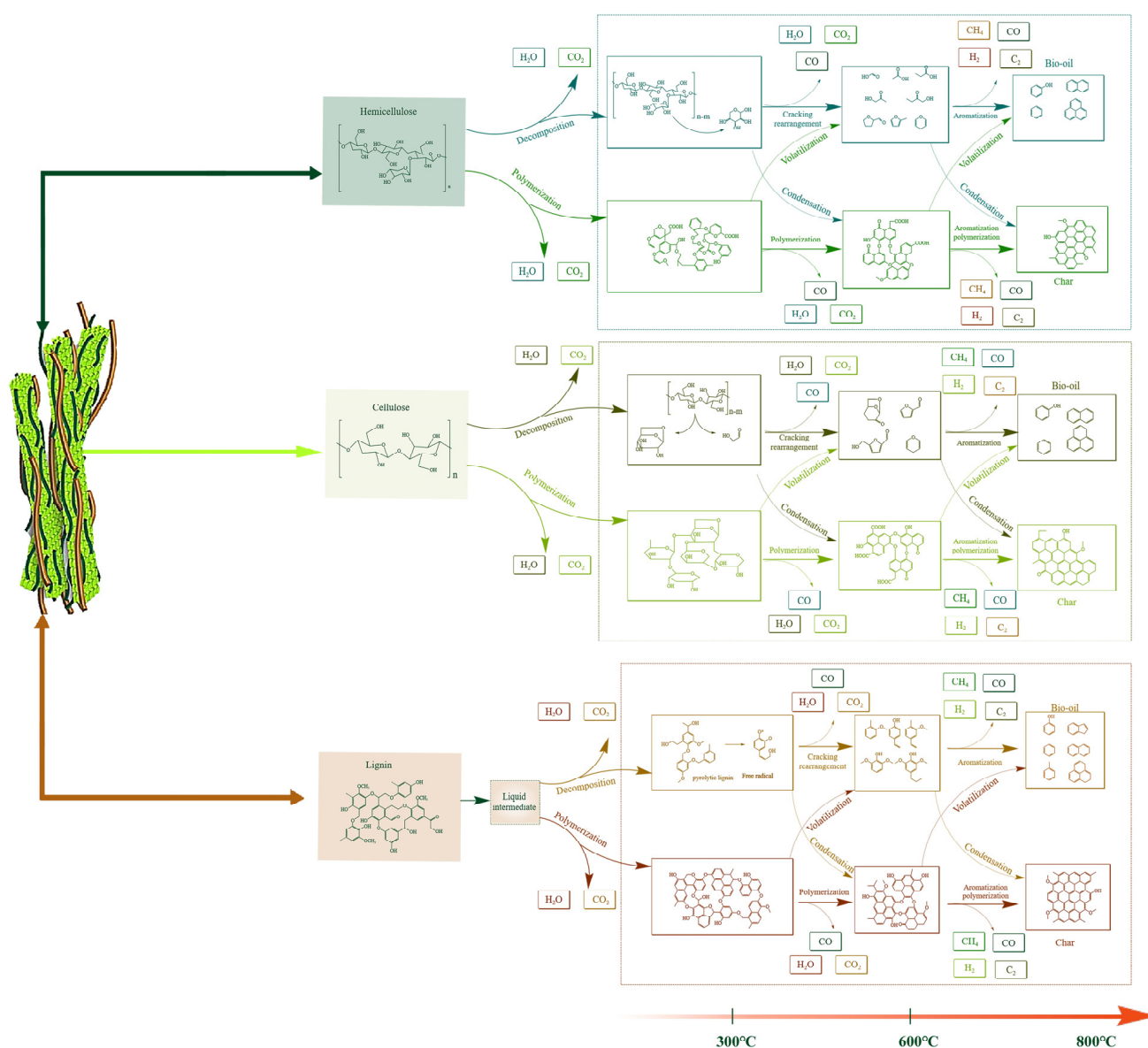


Figure 3. Diagram of pyrolysis mechanism of lignin, cellulose and hemicellulose.

3.2. Liquefaction

Liquefaction is a thermochemical reaction process of biomass feedstock at a medium temperature of 200–400 °C and high pressure of 10–25 MPa, where the feedstock is transformed into bio-oil, gas and solid by cracking and molecular restructuring under the effect of liquefying solvents (water, methanol, ethanol, etc.) [70]. Among them, the bio-oil fraction, as the main product, can reach more than 70%. In the biomass liquefaction process, solvents are crucial in accelerating the decomposition and reaction of biomass and facilitating the production of valuable products such as bio-oil [71,72]. Solvent types are generally divided

into two categories: aqueous solvents and organic solvents [72,73]. Table 4 shows examples of the properties of several commonly used liquefaction solvents, where the solvent vapor pressure, P_c , determines the working pressure of the liquefaction system [74,75]. Compared with other thermochemical utilization technologies, solvent liquefaction offers the following process and product benefits [76–78]: (i) It eliminates the need for pre-drying the feedstock, thus improving the overall process economics. (ii) Solvent liquefaction is more likely to produce high-quality products from catalysts than pyrolysis. (iii) The liquefaction process can be carried out directly without the need for a cumbersome fractionation process. (iv) Solvent liquefaction yields bio-oils with lower moisture and higher calorific value compared to normal pyrolysis [79,80].

Table 4. Properties of commonly used liquefied solvents [72,81].

Solvent		Formula	ρ_c (g/cm ³)	P_c (MPa)	T_c (°C)	Polarity ^c	Dielectric Constant ^d
Water solvent	Water	H ₂ O	0.33	22.10	374	100	78
Organic solvent	Methanol	CH ₄ O	0.27	7.96	240	76.20	32.60
	Ethanol	C ₂ H ₆ O	0.27	6.39	243	65.40	22.40
	Acetone	C ₃ H ₆ O	0.28	4.80	235	35.50	20.40

^c: Water taken as 100; ^d: Determined at 20 °C.

Solvent liquefaction can be categorized into water-based HTL (hydrothermal liquefaction) and organic solvent liquefaction as shown in Table 4 [81]. Solid biopolymers are degraded into smaller molecular fragments to produce bio-oils, bio-based materials and chemicals [82,83]. Around the threshold of 375 °C, 22 MPa, water will become a non-polar solvent and can be used as a reaction medium for organic compounds [84,85]. As can be seen from Table 5, the reaction temperature range of hydrothermal liquefaction is 200–400 °C, and the reaction time is usually within 90 min [86–88], corresponding to the production efficiency is greatly affected by the catalyst, temperature, reaction time, and gasification environment. Organic solvents commonly used for liquefaction include hydrogen donor solvents, diols, glycerol and hydrocarbons [89,90]. Among them, subcritical and supercritical alcohols are widely used in this field due to their unique properties [91]. Some of their main characteristics are as follows. (i) Electrical constants: alcohols have lower electrical constants compared to water, which enables them to better solubilize biomass and facilitate its degradation under ambient and supercritical conditions. (ii) Ease of separation: drying methods using simple alcohols make it easier to separate bio-oils than subcritical and supercritical water, which is very favorable for subsequent processing and purification. (iii) Hydrogen source: Alcohols can increase the yield of bio-oil by providing a source of hydrogen in the reaction process, which helps to inhibit secondary reactions (e.g., re-polymerization and carbonization) [92–94].

In addition, Table 5 summarizes the various liquefaction processes and their production efficiency [95,96]. Synthetic solvent liquefaction using a co-solvent system offers several advantages such as increasing the solubility of the reactants, creating conditions for mild reactions, increasing the yield and obtaining high-quality bio-oil. Microwave-assisted liquefaction is an efficient treatment method that utilizes microwave energy to generate a uniformly distributed heat source in biomass. Microwave heating is faster than traditional heating methods [97]. The plasma in the plasma electrolysis liquefaction technique is a partially ionized gas containing ions, atoms, substrates and free radicals [98]. This technique induces a rapid liquefaction reaction that is not only more efficient but is also environmentally friendly [99].

Table 5. Production of different liquefaction processes and their bio-oil yields.

Liquefaction Technology	Solvent	Reaction Condition	Bio-Oil Yield (wt%)	Elemental Composition (wt%)			HHV (MJ/kg)	Ref.
				C	H	N		
Hydrothermal liquefaction	Water	200–400 °C, <90 min, Air/N ₂ /O ₂ /CO ₂ environment H ₂ pressure	13.9–54.1	66.2–73.6	5.6–7.2	0.1–0.3	24.9–31.9	[29,98,100–102]
Organic solvent liquefaction	Methanol, Ethanol, Polystyrene, Polyethylene, Decalin, Bitumen, Glycerol, Diethylene glycol, Ethylene glycol, Phenol	120–500 °C <60 min Air/N ₂ environment H ₂ pressure	15.1–75.6	50.6–73.9	6.2–8.3	0.1–0.6	20.2–33.2	[81,103–107]
Synthesis solvent liquefaction	Water/Ethanol, Water/Methanol, Water/Glycerol, Glycerol/Methanol	200–340 °C <120 min, Air/N ₂ environment H ₂ pressure	42–67.4	58.4–78.9	5.7–8.1	0.1–0.2	23.1–34.3	[108–111]
Microwave-assisted liquefaction	Glycerol, Methanol, Ethanol, Water, Polyethylene glycol	<200 °C 10–800 W <30 min	90.5	65	6	1	25.1	[112–116]
Plasma electrolytic liquefaction	Polyethylene glycol, Glycerol	<250 °C, <75 min, 100 Hz, <800 V	61.2–97.2	49.5–54.6	8.0–9.6	0.2	20.6–24.7	[117–122]

3.3. Gasification

Gasification is a process in which biomass undergoes an oxidation reaction with a gasifying agent at high temperatures to obtain small molecules of combustible gas [123]. Biomass gasification can be divided into four stages [124–129]. (i) Drying: this is usually carried out at a temperature of 100–200 °C, with low-boiling compounds being evaporated at lower temperatures to increase the quality of the gas produced. (ii) Initial de-volatilization: in the absence of oxidizing agents, initial de-volatilization takes place mainly at 300–900 °C, eventually decomposing into fuel gas, condensable liquids and biochar. (iii) Oxidation: a key process that occurs above 700 °C, where pyrolysis products are oxidized to CO, CO₂ and H₂O under the influence of oxidizing agents (e.g., air, oxygen and steam). (iv) Reduction stage (known as the gasification zone): the coke reacts with the hot gases from the previous zone and is converted to a gaseous form known as syngas.

As shown in Figure 4, biomass reacts with water vapor at high temperatures and pressures to produce syngas (H₂, CO, CO₂, CH₄) and other compounds [130]. Table 6 shows the steam gasification of biomass combustibles and their hydrogen production processes. Steam is generally considered to be an optimum gasification agent to produce rich hydrogen syngas. The gasification of biomass mainly consists of the following reactions:

Carbon gasification: $C + H_2O \rightarrow H_2 + CO$

Water–gas conversion: $CO + H_2O \rightarrow H_2 + CO_2$

Methane reforming: $CH_4 + H_2O \rightarrow 3H_2 + CO$

Hydrocarbon reforming: $C_aH_b + aH_2O \rightarrow aCO + (a + b/2) H_2$

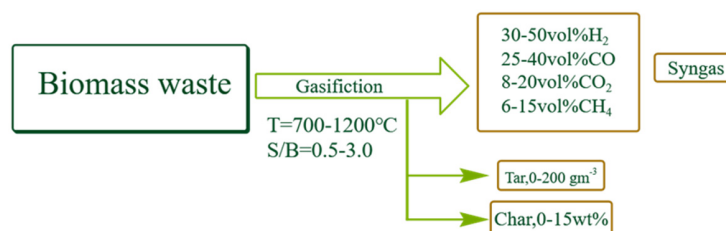
**Figure 4.** Steam gasification of biomass waste.

Table 6. Steam gasification of biomass waste and its hydrogen production.

Feedstock	Operating Conditions	Catalyst	H ₂ Production (wt%) ^e	Gas Yield (Nm ³ /kg)	Tar Content (g/Nm ³)	Ref.
Wood waste	T = 900 °C S/B = 1	Silica sand	5.90	1.22	12.50	[131]
Wood particles	T = 815 °C S/B = 0.84	Olivine	0.76	2.20	4.40	[132]
Almond shells	T = 820 °C S/B = 0.72	Olivine (0.9)/dolomite (0.1)	7.30	1.58	2.13	[133]
Grape	T = 1050 °C S/B = 1.02–3.2	-	1.30	0.60	-	[134]
Palm shells	T = 850 °C S/B = 0.6–1	-	3.70	0.80	1.20	[135]
Sewage sludge	T = 820 °C S/B = 1.5	Silica sand	3.60	0.85	122	[136]
Pine wood	T = 850 °C	-	6.20	1.40	-	[137]
Pine wood particles	T = 820 °C S/B = 2	Silica sand	1.80	0.82	10.30	[138]
Pinewood waste	T = 800–900 °C S/B = 0–2	-	3.30	0.96	142	[139]
Pinewood sawdust	T = 900 °C S/B = 1	Olivine; γ-Al ₂ O ₃	4.40	1.15	22.40	[140,141]

^e: Based on biomass waste.

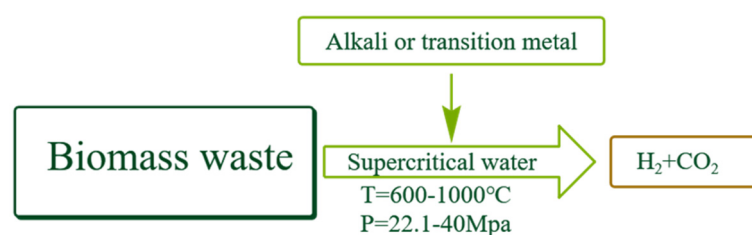
As can be seen in Figure 5, BW undergoes hydrothermal gasification in a liquid water medium [142,143]. In supercritical hydrogasification, water is both the reaction medium and the reactants. As a result, the feedstock does not need to be dried, which largely reduces energy consumption [144]. The following reaction process is included:

Steam reforming reaction: $\text{CH}_n\text{O}_m + (1 - m) \text{H}_2\text{O} \rightarrow (n/2 + 1 - m) \text{H}_2 + \text{CO}$

Water–gas conversion: $\text{CO} + \text{H}_2\text{O} \rightarrow \text{H}_2 + \text{CO}_2$

Methanation: $\text{CO} + 3\text{H}_2 \rightarrow \text{CH}_4 + \text{H}_2\text{O}$

$\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + \text{H}_2\text{O}$

**Figure 5.** Supercritical water gasification of biomass wastes.

Fixed-bed gasifier is one of the traditional gasification technologies with a relatively simple design that enables high carbon conversion, long residence time and low ash carryover. They are usually classified into three types based on the direction of the gas flow, including updraft gasifiers (Figure 6a), downdraft gasifiers (Figure 6b) and crossflow gasifiers (Figure 6c). In addition, there is a fixed-bed gasifier with an open-core gasifier (Figure 6d). Typically, fixed-bed gasifiers are operated at high temperatures around 1000 °C. These gasification systems are used in a wide range of thermal systems from small (10 kW) to large (10 MW).

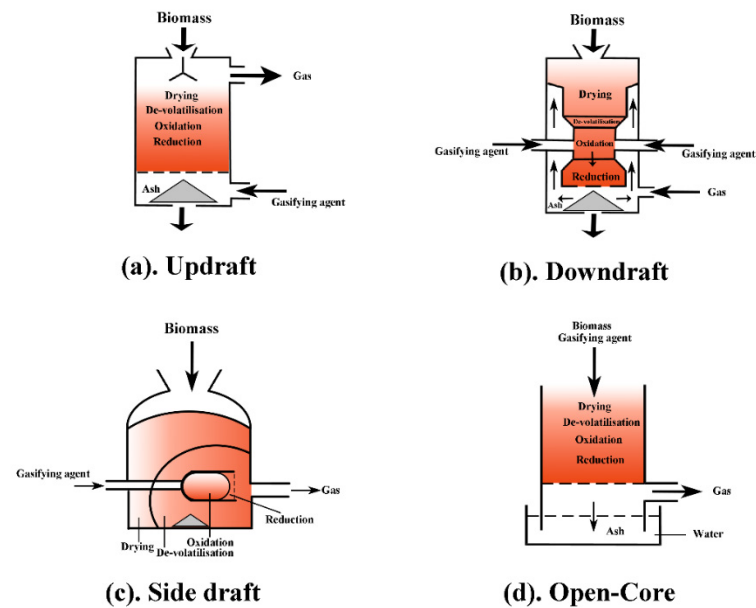


Figure 6. Fixed-bed gasifier, updraft gasifiers (a), downdraft gasifiers (b) and crossflow gasifiers (c). In addition, there is a fixed-bed gasifier with an open-core gasifier (d).

3.4. Combustion

Combustion is the most common technology for the conversion and utilization of BW, where the biomass is fed into a boiler and heat is generated at high temperatures, high pressures and driven by a stream of hot air [145,146]. The combustion process consists of three main steps [147–150]: (i) Drying: The water in the biomass is evaporated at a temperature of about 100–200 °C. (ii) De-volatilization: As the temperature is further increased, the large molecules of organic matter are decomposed into small molecules and volatilized from the detachment, the type and composition of which depends mainly on the heating temperature and residence time. (iii) Volatile combustion: Volatile compounds are further combusted to produce H_2 , CO , CO_2 , CH_4 and coke. The air demand during combustion depends on the nature of the fuel. The excess air ratio ϕ is a vital parameter that affects combustion efficiency. Figure 7 shows the effect of the relative coefficient of the amount of air consumed on the pyrolysis process of biomass at 1 atmospheric pressure. The value of Φ represents the ratio between the real oxygen consumption coefficient and the stoichiometric value throughout the combustion process, i.e., $\Phi = \phi / \phi_c$. In fact, some organic wastes with low calorific value are cost-effective to pyrolyze, and they can be burned with the energy inside of them being converted out directly in the form of heat. The elemental analysis and calorific values of some biomass wastes are shown in Table 7. Aneta [151] studied the combustion and co-combustion characteristics of commonly used biomass fuels, sewage sludge, and coal under different heating rates in an air atmosphere. By employing thermogravimetric analysis and gas composition analysis through mass spectrometry, the combustion characteristics of the fuels were investigated. The results indicated that the co-combustion of biomass and sewage sludge with alternative fuels is a viable and promising waste management alternative. Li et al. [152] studied the high-temperature and rapid combustion of biomass using a CFD combustion model of single-particle biomass. The study reveals the effect that particle size has on the combustion behavior of biomass pellets. As shown in Figure 8, Small pellets of biomass with fast surface heating and short combustion times. Toptas et al. [153] investigated the combustion behavior of biomass-coal mixtures by using the non-isothermal thermogravimetric method. The results showed that the ignition and burnout temperatures of biomass mixed with coal in a 1:1 ratio were lower than those of the pure coal samples. The addition of biomass not only improves the burnout performance of lignite but also serves as an alternative fuel for a coal/biomass co-combustion system.

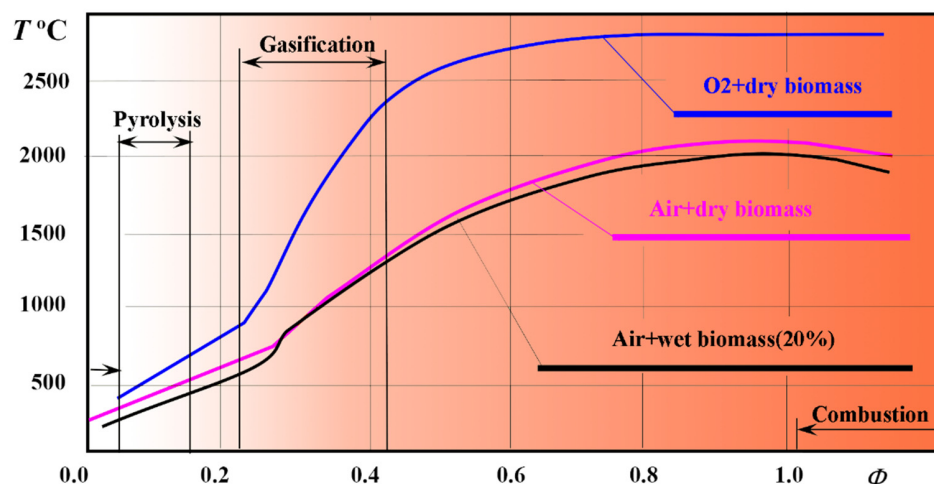


Figure 7. Oxygen consumption during thermochemical conversion of biomass.

Table 7. Elemental analysis of biomass waste and its calorific value.

Feed Stock	Proximate Analysis		Ultimate Analysis					Calorific Value (MJ kg ⁻¹) ^f	Activation Energy (Ea kJ mol ⁻¹)							
	Moisture	Ash	C	H	O	N	S	HHV	α = 0.1	α = 0.2	α = 0.3	α = 0.4	α = 0.5	α = 0.6	α = 0.7	α = 0.8
Wood biomass	2.9	0.3	47.2	6.1	46.3	0.05	0.02	17.59	143.99	177.65	195.06	191.94	188.27	223.07	208.41	129.61
Sewage sludge	5.3	38	30.1	4.6	21.3	4.3	1.2	12.24	29.64	109.16	136.43	150.45	146.80	161.03	122.00	87.32
Oat	7.1	1.5	45.5	6.8	44.0	1.8	0.15	17.20	117.68	184.50	189.66	178.60	199.64	253.49	282.52	164.75
Coal	7.8	4.9	64.2	4.6	22.1	2.2	–	23.53	119.91	145.69	110.19	76.85	53.32	37.67	28.26	23.36

^f: Dry-based water-gas conversion.

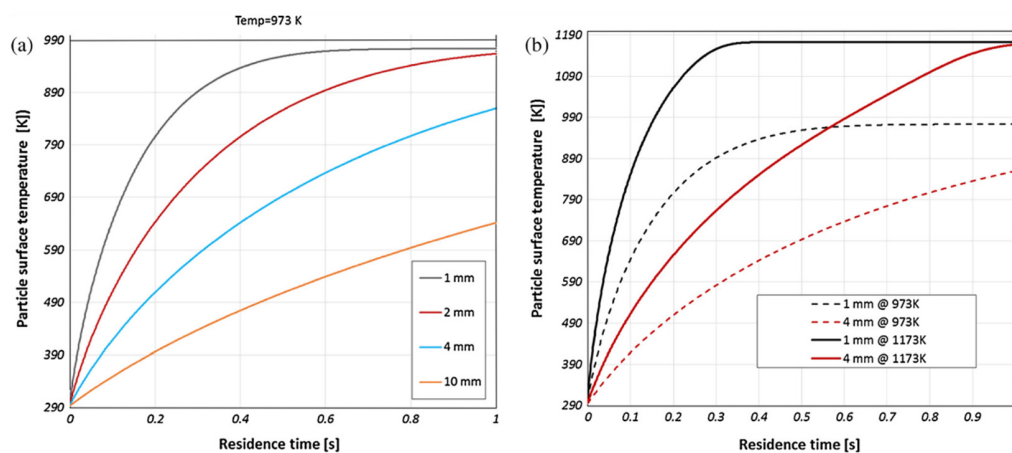


Figure 8. Effect of particle size on combustion, (a) Effect of particle size on surface temperature of biomass particles, (b) Heating history of two different sized particles in the combustion chamber at different temperatures (700 °C and 900 °C) [152].

3.5. By-Products of Biomass Utilization

After processing biomass using the four aforementioned thermal utilization methods, it is inevitable that by-products will be produced alongside the target products. Generally, by-products from biomass utilization can be categorized into three types: solid, liquid, and gas. The main solid by-products are char and ash, which possess high utilization value. When mixed with soil, they can effectively enhance soil fertility and water retention [154]. Vasilatos et al. mixed the ashes from the combustion of olive pomace residues with soil and discovered that the ashes contained significant nutrients such as K, Ca, and Mg, which promote plant growth and successfully improve the quality of exhausted quarries, thus

enhancing the soil matrix [155]. The liquid by-products include tar, aqueous phase liquids (bio-oil), and phenolic compounds. After further refining and processing to remove moisture, bio-oil can be converted into a renewable low-carbon liquid fuel [156–158]. Phenolic resins are important raw materials for many industrial products and have wide applications in food, pharmaceuticals, chemicals, and agriculture. The phenolic substances generated from biomass thermal utilization primarily originate from lignin and cellulose. These phenolic compounds are typically utilized as chemical feedstocks through extraction, ultrasound, or microwave-assisted extraction [159]. Tar may contain toxic components, such as polycyclic aromatic hydrocarbons (PAHs), which pose potential environmental threats. These fractions are often restructured or catalytically converted into gaseous and liquid fuels through high-temperature pyrolysis, which can also be extracted for use as chemical feedstocks [160,161]. The gaseous by-products are more complex, mainly comprising CO_2 , CO , NO_x , SO_x , VOCs and CH_4 . If these gaseous by-products are released freely, they may harm the environment; therefore, in practical applications, catalysts should be employed for catalytic reactions, adsorption, filtration, or chemical absorption to neutralize these gases [162,163].

4. Experimental Analysis

This paper classifies current experimental research into two categories: macroscopic experiments with reactant weights exceeding 10 g and refined experiments with reactant weights below 1 g, according to the extent of product component analysis and the weight of the reactants. The former focuses on collecting gas, liquid, and solid phase components of the heat-treated products and subsequent offline analysis of the desired components, typically the liquid phase. The latter focuses on online analysis of the volatile components of reactants during the heating process.

4.1. Macroscopic Experiment

4.1.1. Fixed-Bed Pyrolysis Technology

Fixed-bed pyrolysis technology is a method of converting BW into various forms of energy and chemicals, with advantages such as high processing capacity, suitability for various types of BW, and low cost [164]. A typical pyrolysis device is shown in Figure 9. In this technology, the feedstock is placed into a fixed-bed reactor and provided with a high-temperature environment through a heater, gradually undergoing pyrolysis under high-temperature anaerobic conditions [165]. This process releases organic matter including solids, liquids, and gases, with the resulting gases and liquids cooled and collected or refined through a condenser to ultimately obtain solid, liquid, and gaseous products. However, biomass fixed-bed pyrolysis technology requires complex operating conditions such as the use of natural gas or BW particles to control temperature and pressure, which increases the difficulty of experimental operation. In addition, the gas and liquid product yields of fixed-bed pyrolysis technology are relatively low and of poor quality [166]. Compared with fixed-bed pyrolysis technology, fluidized-bed pyrolysis technology has advantages in uniform reaction temperature and energy utilization efficiency by utilizing high-speed gas flow to disperse and suspend biomass particles in the air stream.

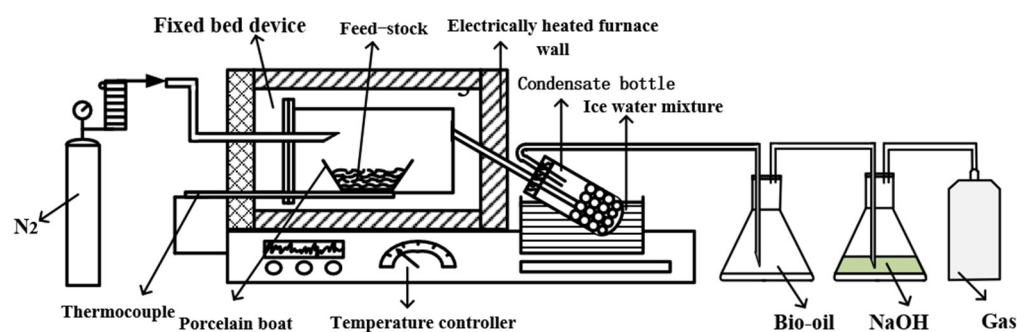


Figure 9. Biomass waste fixed-bed pyrolysis device.

4.1.2. Circulating Fluidized-Bed Pyrolysis Technology

The circulating fluidized-bed (CFB) technology enhances heat and mass transfer efficiency by mixing solid particles (such as coal or biomass) with gas (air or steam) to create a fluidized state within the furnace. This allows solid particles to move freely within the reactor, behaving similarly to a fluid. Furthermore, compared to fixed-bed systems, fluidized beds exhibit a higher tolerance for variations in particle size, density, and shape, eliminating the stringent requirements often associated with fixed beds regarding these physical properties. This characteristic reduces the likelihood of agglomeration and blockage, making it feasible to process fuels with high volatile matter and moisture content effectively within a CFB. The main components of a fluidized bed include the reactor, separator, and circulation system, as illustrated in Figure 10, which depicts a common fluidized-bed system. In summary, compared to fixed-bed technology, CFB technology offers several advantages, including strong adaptability to raw materials, uniform gas–solid mixing, high combustion efficiency, low energy consumption, and favorable environmental performance [167–169].

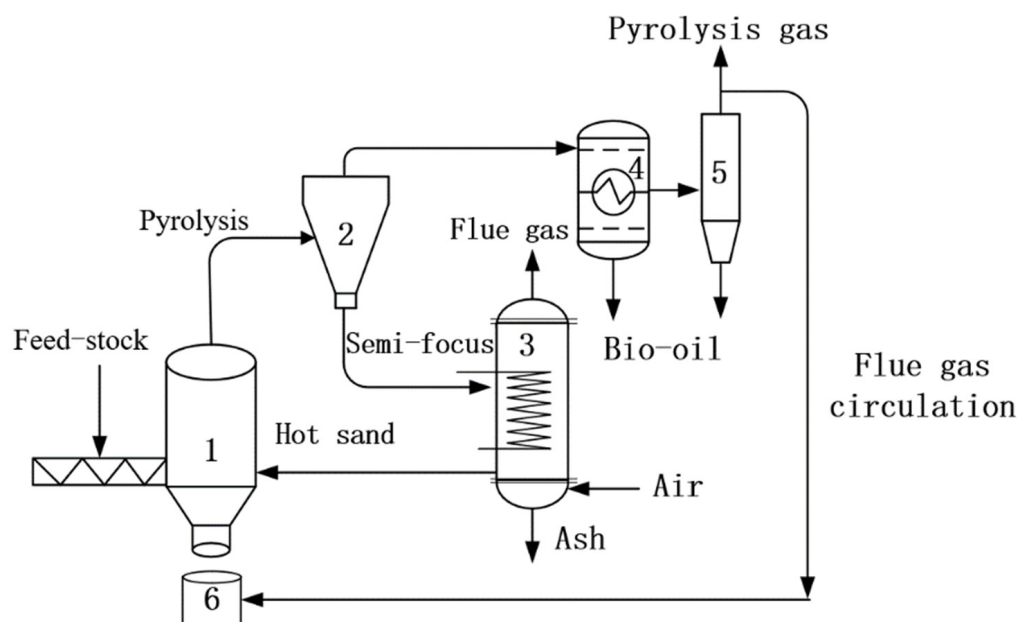


Figure 10. Pyrolysis process diagram of circulating fluidized bed (CFB), 1—pyrolysis chamber, 2—cyclone separator, 3—burner, 4—quencher, 5—electrostatic precipitator, 6—circulating gas heat carrier.

4.1.3. Microwave Pyrolysis Technology

Microwave pyrolysis is to use microwave radiation to make the water molecules in biomass vibrate at high frequency, so that the organic macromolecules are cracked and finally converted into combustible biomass oil, gas and charcoal [170]. Compared with other pyrolysis technologies, microwave heating is faster, has strong penetration, has no need for heating medium, has environmental protection, and is energy-saving [171]. Figure 11 is a schematic diagram of microwave heating and traditional heating methods. In the process of microwave pyrolysis, methods such as rotating transparent glass tubes can be used to improve the uniformity of sample heating [172]. In summary, it is particularly important to select appropriate pyrolysis technologies according to the molecular structure and reaction characteristics of different types of BW to improve pyrolysis efficiency.

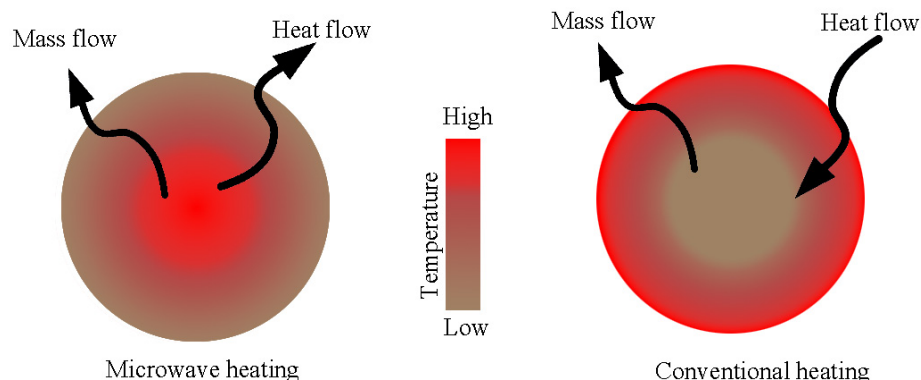


Figure 11. Comparison of microwave heating and traditional heating methods.

4.2. Analysis of Other Pyrolysis Conditions

During biomass pyrolysis experiments, some operating parameters and fuel characteristics can directly or indirectly affect the reaction efficiency of biomass, such as particle size, pyrolysis temperature, heating rate and choice of catalysts can affect the reaction process to a certain extent [173,174].

4.2.1. Particle Size

In the heat processing of BW, the size of the particle plays an important role in thermal efficiency and reaction surface area. Smaller particles promote better transfer of heat and mass, resulting in a more uniform temperature inside the particle. This helps to prevent unnecessary carbonization of the biomass and secondary cracking of the vapor, thus increasing the yield of bio-oil [14]. Table 8 summarizes the effect of different particle sizes on biochar, bio-oil and gas yields. For small particles, the thermal diffusion of internal volatiles is small, but in large particles and large particle sizes, the internal temperature difference leads to more heat transfer and chemical reactions, which significantly affects the distribution of pyrolysis products [175]. Therefore, particle size is a critical parameter in biomass thermal treatment, affecting the efficiency of the reaction and the nature of the products. Xue et al. [176] investigated the pyrolysis characteristics of BW and showed that the larger the particle size, the higher the biochar and pyrolysis gas yields and the lower the bio-oil yield.

Table 8. Effect of particle size on biochar, bio-oil and gas yields.

Feedstock	Pyrolysis Conditions			Particle Size	Yield (wt%)			Ref.
	Temperature	Heating Rate	Reactor		Liquid	Biochar	Gas	
Cotton stalk	550 °C	7 °C min ^{−1}	Fixed bed	0.225–0.425	21	27.5	31.3	[177]
				0.850–1.80	23.8	27.1	28.6	
				0.425–0.850	22	27.8	30.5	
				41.80	22.5	26.6	28.5	
Hazelnut bagasse	500 °C	10 °C min ^{−1}	Fixed bed	0.224–0.425	30.7	27.8	28.5	[178]
				0.425–0.6	33.2	27.9	29	
				0.6–0.85	32.5	27.5	29.7	
				0.85–1.8	31.9	27.4	31	
Rapeseed	550 °C	30 °C min ^{−1}	Fixed bed	<0.425	42.7	23	26	[179]
				0.6–0.85	48	17.5	27.5	
				0.85–1.25	49	16	29	
				1.25–1.8	48	17.5	27.5	
				>1.8	46	21	26.5	
Oil mallee	500 °C	-	Fluidized bed	1–2	52.0	18.8	19.4	[180]
				2.0–3.35	47.5	18.8	28.4	
				4.75–5.6	51.0	20.2	24.0	

Table 8. Cont.

Feedstock	Pyrolysis Conditions			Particle Size	Yield (wt%)			Ref.
	Temperature	Heating Rate	Reactor		Liquid	Biochar	Gas	
Cylindrical wood particles	500 °C	-	Fluidized bed	4	64.2	/	/	[181]
				6	64.6	/	/	
				9	64.0	/	/	
				12	63.0	/	/	
				15	61.2	/	/	

4.2.2. Pyrolysis Temperature

When the temperature is around 300 °C, some of the hemicellulose in the biomass opens up [182]. Around 500 °C, cellulose and most of the lignin have been decomposed, and the biomass is basically carbonized. Therefore, the range of 300 to 500 °C is the main stage of pyrolysis [183]. Both cellulose and lignin can be fully pyrolyzed at around 400 °C, and the pyrolysis is basically complete when the temperature reaches 500 °C [184]. Pyrolysis temperature has a certain impact on the constituents of products. The products of pyrolysis vary with different temperatures [185]. With the increase in temperature, the secondary reaction of volatile matter can be intensified, and the content of volatile matter in the pyrolysis products will increase significantly. The yield of bio-oil increases with the increase in pyrolysis temperature, but when the temperature is too high, the yield of bio-oil decreases [186,187]. As shown in Figure 12, Lewandowski [188] reviewed the changes in CH₄, H₂, and CO yields of coal, willow, and their mixtures at different pyrolysis temperatures.

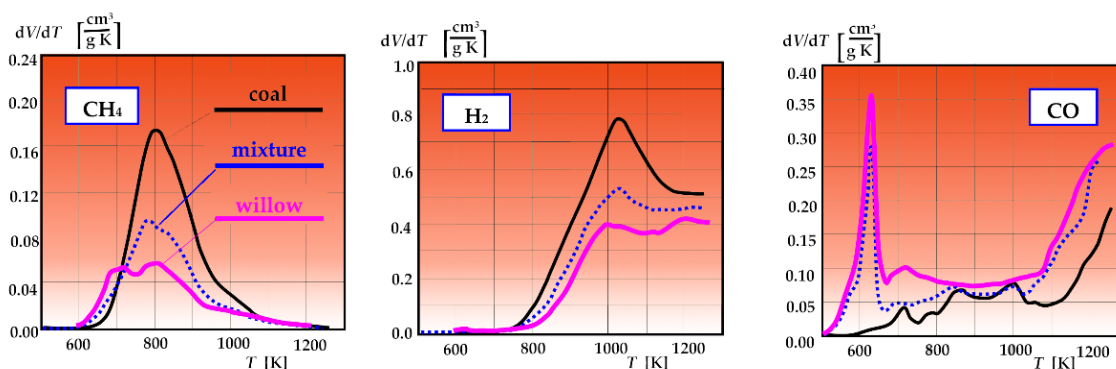


Figure 12. Variation in CH₄, H₂ and CO yields with different temperatures.

4.2.3. Heating Rate

The cellulose pyrolysis products at different heating rates are shown in Figure 13. With slower heating rates, volatiles tend to undergo secondary reactions at higher temperatures, resulting in more biomass being converted to biochar [189]. With a fast heating rate, the pyrolysis reaction of biomass is rapid and the residence time in each temperature interval is short, which reduces the possibility of secondary reactions of volatiles; thus, a fast heating rate is beneficial to increase the yield of gas and bio-oil [190,191]. Brand et al. [192] studied the influence of the heating rate of lignocellulosic biomass during the liquefaction of subcritical water and supercritical ethanol. As can be seen in Figure 14, in liquefaction with subcritical water as solvent, heating rate is a key variable, whereas in supercritical liquefaction with ethanol as solvent, the impact of heating rate is negligible.

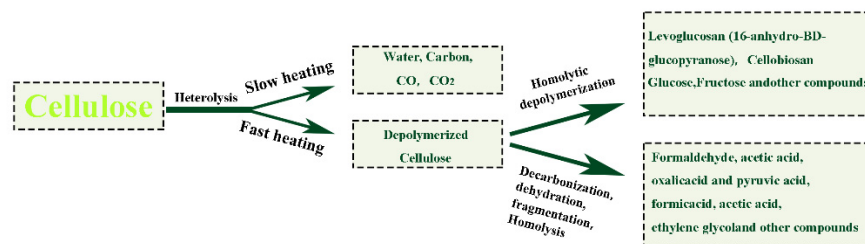


Figure 13. Pyrolysis products of cellulose at different heating rates.

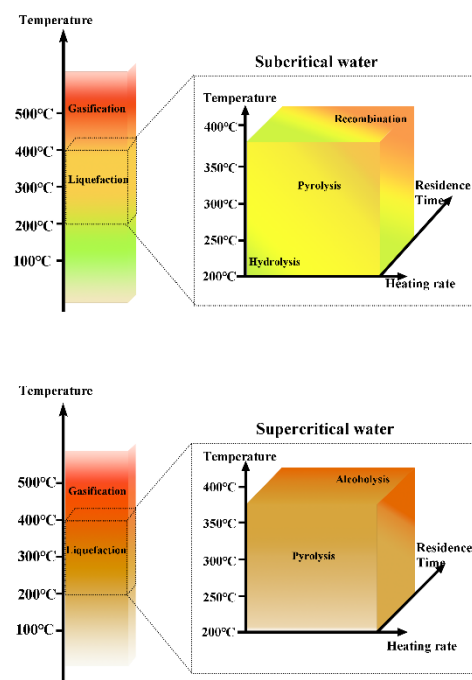


Figure 14. Effect of temperature, heating rate and residence time on biomass.

4.2.4. Catalyst

To increase the yield and quality of bio-oil and to reduce secondary reactions of solid products and volatile components, suitable catalysts are usually added to the pyrolysis process [193]. The type and properties of catalysts directly affect the composition and yield of pyrolysis products. Common catalysts include the following [194–197]: (i) Alkali and alkaline earth metal catalysts, such as sodium, potassium, magnesium and calcium salts. These catalysts are inherently alkaline, which promotes the removal of oxides, reduces the oxygen content in the bio-oil, and improves the acidity of the bio-oil, thereby improving the quality of the bio-oil. (ii) Oxide catalysts, such as CaO, MgO, etc., which can reduce the content of oxygenated compounds such as sugar and acid in the pyrolysis process of biomass and inhibit the secondary reaction in the products, thus improving the yield and quality of bio-oil. (iii) Zeolite catalysts, such as ZSM-5, HZSM-5, USY, MCM-41, etc., which have strong adsorption capacity and can prevent volatiles from polymerization and cooking, thus improving pyrolysis efficiency and product quality. (iv) Nickel-based catalysts, which have high catalytic activity for C-C and C-H bonds in hydrocarbon molecules and can improve the catalytic conversion efficiency of tar and control the gas yield. Therefore, the selection of a suitable catalyst can control the conversion pathway and selectivity of the pyrolysis reaction, which affects the product distribution and quality of biomass pyrolysis [198–200].

In summary, macroscopic experiments can be used to obtain the distribution characteristics of biomass pyrolysis products and to optimize the pyrolysis process and products by adjusting the reaction conditions such as reaction temperature, heating rate and catalyst.

However, to further explore the microscopic mechanism of the reaction process, refined experiments are needed as a supplement.

4.3. Refined Experiments

Refined experiments are an important method for understanding biomass pyrolysis. They can provide fundamental data and theoretical support for optimizing and applying biomass pyrolysis technology by studying the structural changes, chemical reactions, and product distribution of biomass during pyrolysis.

4.3.1. TGA/FTIR/MS Analysis

TGA can obtain curves of sample weight versus time or temperature to obtain key kinetic parameters and is a very useful experimental method for studying the pyrolysis of biomass [201]. Fourier transform infrared spectroscopy and mass spectrometry (FTIR) can analyze the composition and structure of reactants and products to obtain more comprehensive information about the reaction [202]. During biomass pyrolysis, TGA can accurately monitor changes in biomass weight at high temperatures. The FTIR technique determines the changes in biomass species and structure by analyzing the basic composition of reaction products during biomass pyrolysis. The MS can analyze the volatile organic compounds during biomass pyrolysis reactions, and by analyzing the composition and quantity of the products, it can accurately understand the process and structural composition of the biomass reaction products.

Huang et al. [203] investigated the rapid pyrolysis behavior of BW using high heating rate TG-FTIR and Py-GC/MS. Anca et al. [204] performed TGA pyrolysis experiments on pure cellulose and beech wood at different heating rates. The combined use of TGA, FTIR and MS provides a more comprehensive understanding of the mechanism and kinetic behavior of biomass pyrolysis reactions. It is essential for identifying reaction products and determining biomass stability and offers a key reference for rates, experimental design and operating parameters of biomass pyrolysis processes.

4.3.2. Py-GC/MS Analysis

Py-GC/MS is a new analytical technique that allows the rapid heating of samples to high temperatures, separation of the resulting gas-phase components by gas chromatography, and qualitative and quantitative analysis by mass spectrometry to study the composition and structure of the samples [197]. This technique can be used to study the composition and structure of volatile fractions produced by pyrolysis of biomass at different temperature stages [205], and then assess the prospects of high-quality conversion and application of this biomass. Gao et al. [206] investigated the main by-products of biomass pyrolysis using Py-GC-MS. Jiang et al. [207] explored the mechanism of intermolecular interactions during the pyrolysis of BW by Py-GC/MS experiments [208].

In conclusion, biomass pyrolysis is mainly studied by pyrolysis characterization and micro-experimental analyses to investigate the distribution of products and physicochemical properties. However, the existing experimental methods are unable to accurately capture the pyrolysis intermediates and intermediates, and theoretical calculations are needed to assist in revealing the structural properties and formation mechanisms of the intermediates.

5. Theoretical Simulation

5.1. Kinetic Model

Reaction kinetics for biomass pyrolysis models are based on the general global equation of non-homogeneous solid pyrolysis reaction kinetics [208], as follows:



The conversion rate of products can be expressed using the Arrhenius equation [6].

$$\frac{da}{dt} = k(T) \times f(a) = A \exp\left(\frac{-E_a}{RT}\right) \times f(a) \quad (1)$$

where $k(T)$ is the pyrolysis rate constant, a is the conversion rate, t is the conversion time, A is the pre-exponential factor, E_a is the reaction activation energy, R is the universal gas constant and T is the absolute temperature. The degree of conversion (a) is defined as the mass fraction of decomposed solids or released volatiles.

$$\alpha = \frac{m_0 - m}{m_0 - m_f} = \frac{v}{v_f} \quad (2)$$

where m_0 is the initial mass of the solid, m_f is the final mass of the solid, m is the mass of the solid at any given moment, v_f is the mass of volatiles released, and v is the mass of volatiles released at any given moment.

Conventionally, pyrolysis properties of biomass are obtained by isothermal analyses. However, non-isothermal pyrolysis behavior is usually observed during experimental temperature increases. The reaction rate in Equation (1) can be easily translated into a non-isothermal expression that describes the conversion rate as a function of temperature at a constant rate of warming, as follows:

$$\frac{d\alpha}{dT} = \frac{dt}{dT} \frac{d\alpha}{dt} = \frac{1}{\beta} \frac{d\alpha}{dt} \quad (3)$$

where β is a constant heating rate.

Substituting Equation (1) into Equation (3), the non-isothermal rate expression is obtained as:

$$\frac{d\alpha}{dT} = \frac{k(T)}{\beta} \times f(\alpha) = \frac{A}{\beta} \exp\left(\frac{-E_a}{RT}\right) \times f(\alpha) \quad (4)$$

Equations (1) and (4) are chosen based on different reaction mechanisms. The function $f(\alpha)$ can be expressed by more than one equation. The commonly used expression $f(\alpha)$ for biomass pyrolysis is shown in Table 9 [209].

Table 9. The expression of $f(\alpha)$ is based on the common reaction mechanism.

Reaction Mechanism	$f(\alpha)$
Reaction order	
Zero order	1
First order	$(1-\alpha)$
nth order	$(1-\alpha)^n$
Nucleation	
Power law	$n(\alpha)^{(1-1/n)}$
Exponential law	$\ln \alpha$
Avrami–Erofeev	$n(1-\alpha)[- \ln \alpha(1-\alpha)]^{(1-1/n)}$; ($n = 1, 2, 3, 4$)
Prout–Tompkins	$\alpha(1-\alpha)$
Diffusional	
1-Dimensional	$1/2\alpha$
2-Dimensional	$[- \ln \alpha(1-\alpha)]^{-1}$
3-Dimensional (Jander)	$\frac{3}{2}(1-\alpha)^{\frac{2}{3}} \left[1 - (1-\alpha)^{\frac{1}{3}}\right]^{-1}$
3-Dimensional (Ginstling–Brounshtein)	$\frac{3}{2} \left[(1-\alpha)^{-\frac{1}{3}} - 1\right]^{-1}$
Contracting geometry	
Contracting area	$(1-\alpha)^{(1-1/n)}$; $n = 2$
Contracting volume	$(1-\alpha)^{(1-1/n)}$; $n = 3$

5.2. Multi-Scale Computational Fluid Dynamics (CFD) Simulation

In industrial production, the distribution of temperature and fluids within reactors is complicated [210–212]. We simulate these situations with the help of CFD and Reaction Dynamics Modelling, especially during the study of biomass pyrolysis. CFD simulations allow us to carry out the following: (i) Simulate the fluid flow inside the reactor [213]. (ii) Predict the temperature distribution during pyrolysis, both inside the reactor and on the surface of the biomass particles [214]. (iii) Predict the yields and distribution of gaseous, liquid, and solid products under different conditions [215]. These data can be used to adjust the reactor structure, optimize the operating parameters and gain a deeper understanding of the reaction kinetics and pyrolysis mechanism [216]. However, it should be noted that since biomass pyrolysis involves complex physical and chemical changes on multiple scales, the accuracy of CFD simulations is limited by different scale models, as shown in Figure 15. Therefore, the use of multi-scale CFD simulations is an important tool to explore the mechanism of biomass pyrolysis, which is able to describe in detail the flow, heat transfer and reaction processes at different scales [217].

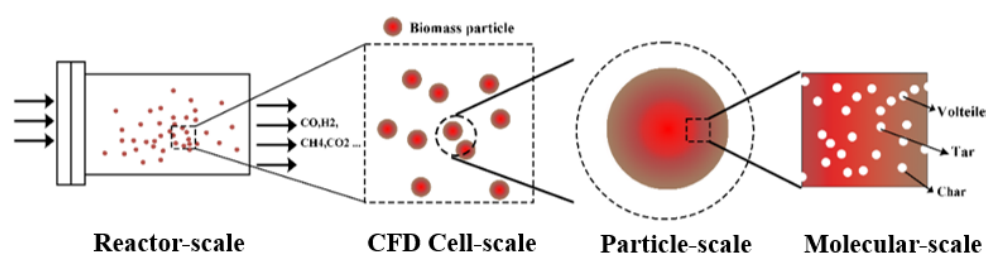


Figure 15. Multi-scale model for CFD simulation.

The reactor scale is a macroscopic scale used to predict the performance of the entire reactor and requires a combination of gas-phase and solid-phase interactions within the reactor, including pyrolysis processes, fluid motion, and heat transfer [218]. In general, the reactor scale takes into account the inlet velocity, reactor pressure and boundary conditions. These conditions can be gained experimentally as well as predicted by means of simulation, which in turn optimizes the reactor parameters [219–221].

In the CFD simulation process, the reactor-scale model undergoes discretization into numerous CFD cells. These cells act as a pivotal link between the particle and reactor scales. By assuming a multiphase flow structure, the CFD cell scale effectively models the thermal, mass transfer, and kinetic phenomena of the gas phase within the reactor. CFD cell scale models are usually classified into homogeneous structure models and inhomogeneous structure models. Homogeneous structure models are primarily utilized to depict the transfer of momentum, heat, and mass between gases and particles, such as the Gidaspow drag model [222], Wen and Yu drag model [223], and Syamlal–O’Brien model [224], etc., which are commonly used in CFD simulations of biomass pyrolysis in fluidized-bed reactors [225–230]. For CFD simulations with larger grids, non-homogeneous structural models are usually considered, which are currently represented by the EMMS model [231–235] and the FTF model [236,237]. These models have been validated in coarse-grid simulations of large fluidized-bed reactors [238,239].

The particle scale is used to study the internal motion, fragmentation and heat transfer behavior of particles during the pyrolysis of biomass. The accuracy of particle-scale models is further improved on the basis of cell-scale models for describing a range of thermal behaviors of biomass particles during pyrolysis. Due to the different particle sizes, the Biot number ($Bi = h_{eff}d/\lambda$) [240], a dimensionless norm, is often defined based on the ratio of the heat transfer thermal resistance inside the particles to that on the particle surface, with the help of which the biomass can be categorized into the thermally thick region ($Bi > 0.1$) and the thermally thin region ($Bi < 0.1$). For biomass particles in the thermally thick region, it is often necessary to use a comprehensive particle conversion model, such as IBM [19], PCM [241], or CUCM [226], to consider the influence of internal particle transport

on conversion rate during the pyrolysis process. Conversely, in the case of biomass particles within the thermally thin region, UCM can effectively portray the reaction kinetics of these particles. In biomass pyrolysis within a fluidized-bed reactor, where particle sizes typically range from 0.1 to 10 mm, the heat transfer dynamics fall within the thermally thick region or during the transition from the thermally thin to the thermally thick region [18,238,240]. Therefore, when performing CFD simulations of biomass pyrolysis in a fluidized-bed reactor, the heat transfer phenomena inside the particles are crucial for analyzing the pyrolysis reaction behavior.

Molecular scale modelling is based on chemical kinetics [11,19,242–244]. One of the commonly used models is SWMM [245]. This model supposes a gradual decomposition of the biomass in a reaction that ultimately produces volatile matter, tar and coke. DAEM, similar to the coal de-volatilization model, has also been applied to biomass pyrolysis [246]. However, the applicability of these models is affected by the type of biomass. To address this problem, CSBM was developed, which was designed to forecast the pyrolysis of large molecular fractions such as cellulose, hemicellulose and lignin [12,211,247]. By combining data on the macromolecular components of different feedstocks, the pyrolysis behavior of different biomasses can be described.

A multi-scale coupled model, as shown in Figure 16, is often required in practical CFD simulations, which can provide a more comprehensive understanding of the biomass pyrolysis process [248]. For example, information on reaction kinetics obtained from the molecular scale can be used to determine model parameters at the particle scale, and CFD cell-scale simulations can provide information on fluid dynamics, which can ultimately be used for reactor-scale performance prediction [249,250]. This multi-scale coupled approach helps to bridge the gap between models at different scales and improves the accuracy and reliability of simulations.

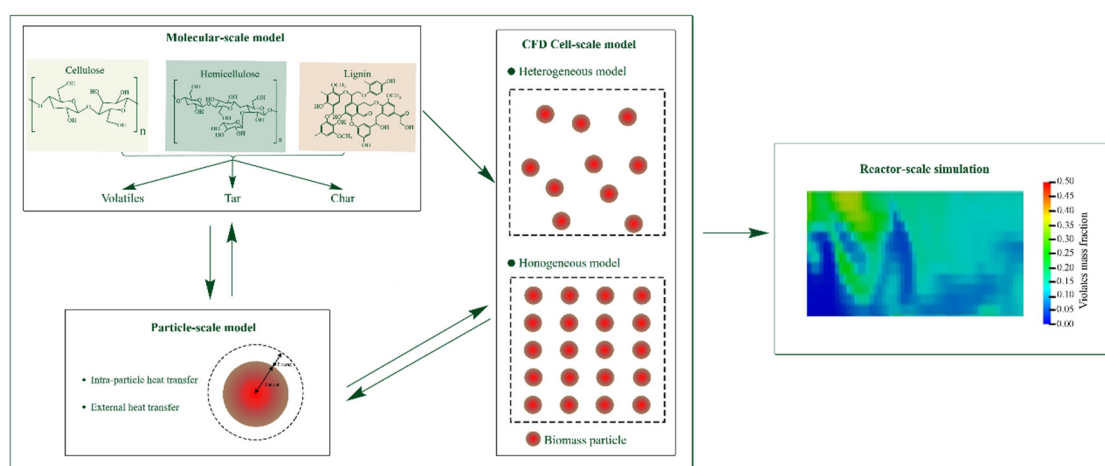


Figure 16. Multi-scale model coupling for CFD simulation.

6. Conclusions and Prospects

This article summarizes the methods and current status of BW utilization and explores future research directions. It discusses various aspects of the BW decomposition process, from microscopic experiments to macroscopic analyses, to deepen our understanding of its structure and utilization methods. Although significant achievements have been made since the introduction of BW utilization, many countries around the world have begun to adopt BW as a new energy source to replace fossil fuels. However, the energy conversion efficiency of BW remains relatively low; therefore, appropriate catalysts should be introduced during the thermal utilization of BW to enhance conversion efficiency, and the by-products generated in the process should be recycled to fully improve the utilization rate of the raw materials. Furthermore, to achieve sustainable development, carbon capture and waste gas treatment technologies should be widely integrated into the thermal utilization of BW in the future. While the value of BW utilization has been validated,

achieving low-cost industrial production remains a significant challenge in this field. The complexity and diversity of BW increase the cost and complexity of pyrolysis reactors. Consequently, future research should focus on the kinetic analysis of biomass and the multiphase flow simulations based on CFD at the unit scale. Utilizing artificial intelligence technology can help construct more precise theoretical models at the microscopic scale and integrate them with macroscopic reaction kinetics models to achieve faster and more effective pyrolysis optimization.

Author Contributions: Conceptualization, B.C. and Y.G.; writing—original draft preparation, J.Z. and N.C.; writing—review and editing, J.Z.; visualization, J.Z. and N.C.; project administration, Y.G.; funding acquisition, Y.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by the Project of Shanghai Municipal Science and Technology Commission, (Grant No. 19195810800), and the National Natural Science Foundation of China, (Grant No. 51606116).

Data Availability Statement: No data were used for the research described in the article.

Conflicts of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Nomenclature

BW	Biomass waste
TGA	Thermogravimetric analyzer
CFD	Computational Fluid Dynamics
HHV	High heating value
Py-GC/MS	Pyrolysis gas chromatography mass spectrometry
SCW	Supercritical water gasification
HTL	Hydrothermal liquefaction
CFB	Circulating fluidized bed
FTIR	Fourier transform infrared spectroscopy
MS	Mass spectrometry
FTF	Filtered Two-Fluid
EMMS	Energy-Minimization Multi-Scale
SWMM	Stepwise Mechanism Model
DAEM	Distributed Activation Energy Model
CSBM	Chemistry Structure-Based Model
IBM	Interface-based model
PCM	Progressive conversion model
CUCM	Corrected uniform conversion model
UCM	Uniform conversion model

References

1. Cai, W.; Luo, Z.; Zhou, J.; Wang, Q. A review on the selection of raw materials and reactors for biomass fast pyrolysis in China. *Fuel Process. Technol.* **2021**, *221*, 106919. [[CrossRef](#)]
2. Cong, H.; Meng, H.; Mašek, O.; Yao, Z.; Li, L.; Yu, B.; Qin, C.; Zhao, L. Comprehensive analysis of industrial-scale heating plants based on different biomass slow pyrolysis technologies: Product property, energy balance, and ecological impact. *Clean. Eng. Technol.* **2022**, *6*, 100391. [[CrossRef](#)]
3. Kumar Sarangi, P.; Subudhi, S.; Bhatia, L.; Saha, K.; Mudgil, D.; Prasad Shadangi, K.; Srivastava, R.K.; Pattnaik, B.; Arya, R.K. Utilization of agricultural waste biomass and recycling toward circular bioeconomy. *Environ. Sci. Pollut. Res.* **2023**, *30*, 8526–8539. [[CrossRef](#)] [[PubMed](#)]
4. Yogalakshmi, K.N.; Poornima, D.T.; Sivashanmugam, P.; Kavitha, S.; Yakesh, K.R.; Varjani, S.; AdishKumar, S.; Kumar, G.; Rajesh, B.J. Lignocellulosic biomass-based pyrolysis: A comprehensive review. *Chemosphere* **2022**, *286 Pt 2*, 131824.
5. Hoang, A.T.; Ong, H.C.; Fattah, I.M.R.; Chong, C.T.; Cheng, C.K.; Sakthivel, R.; Ok, Y.S. Progress on the lignocellulosic biomass pyrolysis for biofuel production toward environmental sustainability. *Fuel Process. Technol.* **2021**, *223*, 106997. [[CrossRef](#)]
6. Bai, X.; Wang, G.; Yu, Y.; Wang, D.; Wang, Z. Changes in the physicochemical structure and pyrolysis characteristics of wheat straw after rod-milling pretreatment. *Bioresour. Technol.* **2018**, *250*, 770–776. [[CrossRef](#)]

7. Lungkadee, T.; Tipayawong, K.Y.; Tangparitkul, S.; Jaroenkhasemmesuk, C.; Chaichana, C.; Wongspai, W.; Tipayawong, N. Decarbonizing a Thai Coal Power Plant: Effect of Flue Gas Loads on Carbon Capture Performance and Economics. *Appl. Eng. Lett.* **2024**, *9*, 12–21. [\[CrossRef\]](#)
8. Chen, B.; Yang, Y. Co-pyrolysis of algae with other carbon-based materials: A review on synergistic products and molecular reaction pathway. *J. Therm. Anal. Calorim.* **2023**, *148*, 2233–2249. [\[CrossRef\]](#)
9. Elqadhi, M.; Skrbic, S.; Mohamoud, O.; Asonja, A. Energy integration of corn cob in the process of drying the corn seeds. *Therm. Sci.* **2024**, *28 Pt B*, 3325–3336. [\[CrossRef\]](#)
10. Mani, T.; Murugan, P.; Abedi, J.; Mahinpey, N. Pyrolysis of wheat straw in a thermogravimetric analyzer: Effect of particle size and heating rate on devolatilization and estimation of global kinetics. *Chem. Eng. Res. Des.* **2010**, *88*, 952–958. [\[CrossRef\]](#)
11. Anca-Couce, A. Reaction mechanisms and multi-scale modelling of lignocellulosic biomass pyrolysis. *Prog. Energy Combust. Sci.* **2016**, *53*, 41–79. [\[CrossRef\]](#)
12. Anca-Couce, A.; Obernberger, I. Application of a detailed biomass pyrolysis kinetic scheme to hardwood and softwood torrefaction. *Fuel* **2016**, *167*, 158–167. [\[CrossRef\]](#)
13. Boateng, A.A.; Mtui, P.L. CFD modeling of space-time evolution of fast pyrolysis products in a bench-scale fluidized-bed reactor. *Appl. Therm. Eng.* **2012**, *33–34*, 190–198. [\[CrossRef\]](#)
14. Zhang, Y.; Liang, Y.; Li, S.; Yuan, Y.; Zhang, D.; Wu, Y.; Xie, H.; Brindhadevi, K.; Pugazhendhi, A.; Xia, C. A review of biomass pyrolysis gas: Forming mechanisms, influencing parameters, and product application upgrades. *Fuel* **2023**, *347*, 128461. [\[CrossRef\]](#)
15. Kan, T.; Strezov, V.; Evans, T.J. Lignocellulosic biomass pyrolysis: A review of product properties and effects of pyrolysis parameters. *Renew. Sustain. Energy Rev.* **2016**, *57*, 1126–1140. [\[CrossRef\]](#)
16. Gao, Y.; Xu, C.; Cui, D.; Rout, L.; Ding, K.; Shi, L.; Zhang, S.; Lv, P.; Li, B.; Yu, G.; et al. Decoupling study on the influence of the interaction between biomass hydrochar and coal during co-pyrolysis on the char structure evolution. *Renew. Energy* **2024**, *231*, 120938. [\[CrossRef\]](#)
17. Tchapda, A.; Pisupati, S. A Review of Thermal Co-Conversion of Coal and Biomass/Waste. *Energies* **2014**, *7*, 1098–1148. [\[CrossRef\]](#)
18. Gómez-Barea, A.; Leckner, B. Modeling of biomass gasification in fluidized bed. *Prog. Energy Combust. Sci.* **2010**, *36*, 444–509. [\[CrossRef\]](#)
19. Haberer, I.; Skreiberg, Ø.; Lazar, J.; Haugen, N.E.L. Numerical models for thermochemical degradation of thermally thick woody biomass, and their application in domestic wood heating appliances and grate furnaces. *Prog. Energy Combust. Sci.* **2017**, *63*, 204–252. [\[CrossRef\]](#)
20. Brojanigo, S.; Gronchi, N.; Cazzorla, T.; Wong, T.S.; Basaglia, M.; Favaro, L.; Casella, S. Engineering Cupriavidus necator DSM 545 for the one-step conversion of starchy waste into polyhydroxyalkanoates. *Bioresour. Technol.* **2022**, *347*, 126383. [\[CrossRef\]](#)
21. Huang, J.; Qiao, Y.; Wei, X.; Zhou, J.; Yu, Y.; Xu, M. Effect of torrefaction on steam gasification of starchy food waste. *Fuel* **2019**, *253*, 1556–1564. [\[CrossRef\]](#)
22. Aljabri, H.; Das, P.; Khan, S.; AbdulQuadir, M.; Thaher, M.; Hawari, A.H.; Al-Shamary, N.M. A study to investigate the energy recovery potential from different macromolecules of a low-lipid marine *Tetraselmis* sp. biomass through HTL process. *Renew. Energy* **2022**, *189*, 78–89. [\[CrossRef\]](#)
23. Olguin, E.J.; Castillo, O.S.; Mendoza, A.; Tapia, K.; Gonzalez-Portela, R.E.; Hernandez-Landa, V.J. Dual purpose system that treats anaerobic effluents from pig waste and produce *Neochloris oleoabundans* as lipid rich biomass. *New Biotechnol.* **2015**, *32*, 387–395. [\[CrossRef\]](#) [\[PubMed\]](#)
24. Mishra, S.; Mohanty, K. Comprehensive characterization of microalgal isolates and lipid-extracted biomass as zero-waste bioenergy feedstock: An integrated bioremediation and biorefinery approach. *Bioresour. Technol.* **2019**, *273*, 177–184. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Dai, L.; Wang, Y.; Liu, Y.; He, C.; Ruan, R.; Yu, Z.; Jiang, L.; Zeng, Z.; Wu, Q. A review on selective production of value-added chemicals via catalytic pyrolysis of lignocellulosic biomass. *Sci. Total Environ.* **2020**, *749*, 142386. [\[CrossRef\]](#)
26. Di Blasi, C.; Branca, C.; Galgano, A. Biomass Screening for the Production of Furfural via Thermal Decomposition. *Ind. Eng. Chem. Res.* **2010**, *49*, 2658–2671. [\[CrossRef\]](#)
27. Branca, C.; Di Blasi, C.; Galgano, A. Pyrolysis of Corncobs Catalyzed by Zinc Chloride for Furfural Production. *Ind. Eng. Chem. Res.* **2010**, *49*, 9743–9752. [\[CrossRef\]](#)
28. Demirbaş, A. Thermochemical Conversion of Biomass to Liquid Products in the Aqueous Medium. *Energy Sources* **2005**, *27*, 1235–1243. [\[CrossRef\]](#)
29. de Caprariis, B.; De Filippis, P.; Petrullo, A.; Scarsella, M. Hydrothermal liquefaction of biomass: Influence of temperature and biomass composition on the bio-oil production. *Fuel* **2017**, *208*, 618–625. [\[CrossRef\]](#)
30. Rathnayake, M.; Chaireongsirikul, T.; Svangariyaskul, A.; Lawtrakul, L.; Toochinda, P. Process simulation based life cycle assessment for bioethanol production from cassava, cane molasses, and rice straw. *J. Clean. Prod.* **2018**, *190*, 24–35. [\[CrossRef\]](#)
31. Ruiz-Aquino, F.; González-Peña, M.M.; Valdez-Hernández, J.I.; Revilla, U.S.; Romero-Manzanares, A. Chemical characterization and fuel properties of wood and bark of two oaks from Oaxaca, Mexico. *Ind. Crops Prod.* **2015**, *65*, 90–95. [\[CrossRef\]](#)
32. Intani, K.; Latif, S.; Kabir, A.K.M.R.; Müller, J. Effect of self-purging pyrolysis on yield of biochar from maize cobs, husks and leaves. *Bioresour. Technol.* **2016**, *218*, 541–551. [\[CrossRef\]](#) [\[PubMed\]](#)

33. Irfan, M.; Chen, Q.; Yue, Y.; Pang, R.; Lin, Q.; Zhao, X.; Chen, H. Co-production of biochar, bio-oil and syngas from halophyte grass (*Achnatherum splendens* L.) under three different pyrolysis temperatures. *Bioresour. Technol.* **2016**, *211*, 457–463. [\[CrossRef\]](#) [\[PubMed\]](#)
34. Lee, X.J.; Lee, L.Y.; Gan, S.; Thangalazhy-Gopakumar, S.; Ng, H.K. Biochar potential evaluation of palm oil wastes through slow pyrolysis: Thermochemical characterization and pyrolytic kinetic studies. *Bioresour. Technol.* **2017**, *236*, 155–163. [\[CrossRef\]](#) [\[PubMed\]](#)
35. He, Q.; Guo, Q.; Ding, L.; Wei, J.; Yu, G. CO₂ gasification of char from raw and torrefied biomass: Reactivity, kinetics and mechanism analysis. *Bioresour. Technol.* **2019**, *293*, 122087. [\[CrossRef\]](#)
36. Rago, Y.P.; Surroop, D.; Mohee, R. Assessing the potential of biofuel (biochar) production from food wastes through thermal treatment. *Bioresour. Technol.* **2018**, *248*, 258–264. [\[CrossRef\]](#)
37. Li, W.; Dang, Q.; Brown, R.C.; Laird, D.; Wright, M.M. The impacts of biomass properties on pyrolysis yields, economic and environmental performance of the pyrolysis-bioenergy-biochar platform to carbon negative energy. *Bioresour. Technol.* **2017**, *241*, 959–968. [\[CrossRef\]](#)
38. Kumar, M.; Shukla, S.K.; Upadhyay, S.N.; Mishra, P.K. Analysis of thermal degradation of banana (*Musa balbisiana*) trunk biomass waste using iso-conversional models. *Bioresour. Technol.* **2020**, *310*, 123393. [\[CrossRef\]](#)
39. Zhang, Y.; Yao, A.; Song, K. Torrefaction of cultivation residue of *Auricularia auricula-judae* to obtain biochar with enhanced fuel properties. *Bioresour. Technol.* **2016**, *206*, 211–216. [\[CrossRef\]](#)
40. Zheng, H.; Guo, W.; Li, S.; Chen, Y.; Wu, Q.; Feng, X.; Yin, R.; Ho, S.-H.; Ren, N.; Chang, J.-S. Adsorption of p-nitrophenols (PNP) on microalgal biochar: Analysis of high adsorption capacity and mechanism. *Bioresour. Technol.* **2017**, *244*, 1456–1464. [\[CrossRef\]](#)
41. Abdullahi, Z.H.; Marselin, F.N.; Khaironizam, N.I.A.; Fauzi, N.F.A.; Wan Maznah, W.O. Growth stage-related biomass, pigments, and biochemical composition of *Stichococcus bacillaris*, *Synechococcus* sp. and *Trentepohlia aurea* isolated from Gua Tempurung, a cave in Malaysia. *Plant Physiol. Biochem.* **2023**, *197*, 107633. [\[CrossRef\]](#) [\[PubMed\]](#)
42. Zhang, Y.; Ding, Z.; Shahadat Hossain, M.; Maurya, R.; Yang, Y.; Singh, V.; Kumar, D.; Salama, E.S.; Sun, X.; Sindhu, R.; et al. Recent advances in lignocellulosic and algal biomass pretreatment and its biorefinery approaches for biochemicals and bioenergy conversion. *Bioresour. Technol.* **2023**, *367*, 128281. [\[CrossRef\]](#) [\[PubMed\]](#)
43. Manikandan, S.; Vickram, S.; Sirohi, R.; Subbaiya, R.; Krishnan, R.Y.; Karmegam, N.; Sumathijones, C.; Rajagopal, R.; Chang, S.W.; Ravindran, B.; et al. Critical review of biochemical pathways to transformation of waste and biomass into bioenergy. *Bioresour. Technol.* **2023**, *372*, 128679. [\[CrossRef\]](#) [\[PubMed\]](#)
44. Yan, P.; Wang, H.; Liao, Y.; Wang, C. Zeolite catalysts for the valorization of biomass into platform compounds and biochemicals/biofuels: A review. *Renew. Sustain. Energy Rev.* **2023**, *178*, 113219. [\[CrossRef\]](#)
45. Guo, G.; Zhang, K.; Liu, C.; Xie, S.; Li, X.; Li, B.; Shu, J.; Niu, Y.; Zhu, H.; Ding, M.; et al. Comparative investigation on thermal decomposition of powdered and pelletized biomasses: Thermal conversion characteristics and apparent kinetics. *Bioresour. Technol.* **2020**, *301*, 122732. [\[CrossRef\]](#)
46. Papari, S.; Hawboldt, K. A review on the pyrolysis of woody biomass to bio-oil: Focus on kinetic models. *Renew. Sustain. Energy Rev.* **2015**, *52*, 1580–1595. [\[CrossRef\]](#)
47. Kong, Z.; Liaw, S.B.; Gao, X.; Yu, Y.; Wu, H. Leaching characteristics of inherent inorganic nutrients in biochars from the slow and fast pyrolysis of mallee biomass. *Fuel* **2014**, *128*, 433–441. [\[CrossRef\]](#)
48. Neves, D.; Thunman, H.; Matos, A.; Tarelho, L.; Gómez-Barea, A. Characterization and prediction of biomass pyrolysis products. *Prog. Energy Combust. Sci.* **2011**, *37*, 611–630. [\[CrossRef\]](#)
49. Balagurumurthy, B.; Bhaskar, T. Hydropyrolysis of lignocellulosic biomass: State of the art review. *Biomass Convers. Biorefinery* **2013**, *4*, 67–75. [\[CrossRef\]](#)
50. Tripathi, M.; Sahu, J.N.; Ganesan, P. Effect of process parameters on production of biochar from biomass waste through pyrolysis: A review. *Renew. Sustain. Energy Rev.* **2016**, *55*, 467–481. [\[CrossRef\]](#)
51. Thangalazhy-Gopakumar, S.; Al-Nadheri, W.M.A.; Jegarajan, D.; Sahu, J.N.; Mubarak, N.M.; Nizamuddin, S. Utilization of palm oil sludge through pyrolysis for bio-oil and bio-char production. *Bioresour. Technol.* **2015**, *178*, 65–69. [\[CrossRef\]](#) [\[PubMed\]](#)
52. Liu, S.; Lu, H.; Hu, R.; Shupe, A.; Lin, L.; Liang, B. A sustainable woody biomass biorefinery. *Biotechnol. Adv.* **2012**, *30*, 785–810. [\[CrossRef\]](#) [\[PubMed\]](#)
53. Chen, Y.-D.; Liu, F.; Ren, N.-Q.; Ho, S.-H. Revolutions in algal biochar for different applications: State-of-the-art techniques and future scenarios. *Chin. Chem. Lett.* **2020**, *31*, 2591–2602. [\[CrossRef\]](#)
54. Lehto, J.; Oasmaa, A.; Solantausta, Y.; Kytö, M.; Chiaramonti, D. Review of fuel oil quality and combustion of fast pyrolysis bio-oils from lignocellulosic biomass. *Appl. Energy* **2014**, *116*, 178–190. [\[CrossRef\]](#)
55. Torri, I.D.V.; Paasikallio, V.; Faccini, C.S.; Huff, R.; Caramão, E.B.; Sacon, V.; Oasmaa, A.; Zini, C.A. Bio-oil production of softwood and hardwood forest industry residues through fast and intermediate pyrolysis and its chromatographic characterization. *Bioresour. Technol.* **2016**, *200*, 680–690. [\[CrossRef\]](#)
56. Mulligan, C.J.; Strezov, L.; Strezov, V. Thermal Decomposition of Wheat Straw and Mallee Residue Under Pyrolysis Conditions. *Energy Fuels* **2010**, *24*, 46–52. [\[CrossRef\]](#)
57. Grierson, S.; Strezov, V.; Ellem, G.; McGregor, R.; Herbertson, J. Thermal characterisation of microalgae under slow pyrolysis conditions. *J. Anal. Appl. Pyrolysis* **2009**, *85*, 118–123. [\[CrossRef\]](#)

58. Bates, R.B.; Ghoniem, A.F. Biomass torrefaction: Modeling of volatile and solid product evolution kinetics. *Bioresour. Technol.* **2012**, *124*, 460–469. [\[CrossRef\]](#)
59. Shankar Tumuluru, J.; Sokhansanj, S.; Hess, J.R.; Wright, C.T.; Boardman, R.D. REVIEW: A review on biomass torrefaction process and product properties for energy applications. *Ind. Biotechnol.* **2011**, *7*, 384–401. [\[CrossRef\]](#)
60. Acharya, B.; Sule, I.; Dutta, A. A review on advances of torrefaction technologies for biomass processing. *Biomass Convers. Biorefinery* **2012**, *2*, 349–369. [\[CrossRef\]](#)
61. Mohan, D.; Pittman, C.U.; Steele, P.H. Pyrolysis of Wood/Biomass for Bio-oil: A Critical Review. *Energy Fuels* **2006**, *20*, 848–889. [\[CrossRef\]](#)
62. Ahmad, M.; Rajapaksha, A.U.; Lim, J.E.; Zhang, M.; Bolan, N.; Mohan, D.; Vithanage, M.; Lee, S.S.; Ok, Y.S. Biochar as a sorbent for contaminant management in soil and water: A review. *Chemosphere* **2014**, *99*, 19–33. [\[CrossRef\]](#) [\[PubMed\]](#)
63. García-Pérez, M.; Chaala, A.; Pakdel, H.; Kretschmer, D.; Roy, C. Vacuum pyrolysis of softwood and hardwood biomass. *J. Anal. Appl. Pyrolysis* **2007**, *78*, 104–116. [\[CrossRef\]](#)
64. Bridgwater, A.V. Review of fast pyrolysis of biomass and product upgrading. *Biomass Bioenergy* **2012**, *38*, 68–94. [\[CrossRef\]](#)
65. Zheng, M.; Wang, Z.; Li, X.; Qiao, X.; Song, W.; Guo, L. Initial reaction mechanisms of cellulose pyrolysis revealed by ReaxFF molecular dynamics. *Fuel* **2016**, *177*, 130–141. [\[CrossRef\]](#)
66. Bridgwater, A.V.; Meier, D.; Radlein, D. An overview of fast pyrolysis of biomass. *Org. Geochem.* **1999**, *30*, 1479–1493. [\[CrossRef\]](#)
67. Li, C.; Zhao, X.; Wang, A.; Huber, G.W.; Zhang, T. Catalytic Transformation of Lignin for the Production of Chemicals and Fuels. *Chem. Rev.* **2015**, *115*, 11559–11624. [\[CrossRef\]](#)
68. Liu, C.; Hu, J.; Zhang, H.; Xiao, R. Thermal conversion of lignin to phenols: Relevance between chemical structure and pyrolysis behaviors. *Fuel* **2016**, *182*, 864–870. [\[CrossRef\]](#)
69. Li, Y.; Cui, D.; Tong, Y.; Xu, L. Study on structure and thermal stability properties of lignin during thermostabilization and carbonization. *Int. J. Biol. Macromol.* **2013**, *62*, 663–669. [\[CrossRef\]](#)
70. Huang, H.-J.; Yuan, X.-Z. Recent progress in the direct liquefaction of typical biomass. *Prog. Energy Combust. Sci.* **2015**, *49*, 59–80. [\[CrossRef\]](#)
71. Huber, G.W.; Iborra, S.; Corma, A. Synthesis of Transportation Fuels from Biomass: Chemistry, Catalysts, and Engineering. *Chem. Rev.* **2006**, *106*, 4044–4098. [\[CrossRef\]](#) [\[PubMed\]](#)
72. Mazaheri, H.; Lee, K.T.; Bhatia, S.; Mohamed, A.R. Sub/supercritical liquefaction of oil palm fruit press fiber for the production of bio-oil: Effect of solvents. *Bioresour. Technol.* **2010**, *101*, 7641–7647. [\[CrossRef\]](#) [\[PubMed\]](#)
73. Nanda, S.; Rana, R.; Hunter, H.N.; Fang, Z.; Dalai, A.K.; Kozinski, J.A. Hydrothermal catalytic processing of waste cooking oil for hydrogen-rich syngas production. *Chem. Eng. Sci.* **2019**, *195*, 935–945. [\[CrossRef\]](#)
74. Peterson, A.A.; Vogel, F.; Lachance, R.P.; Fröling, M.; Antal, J.M.J.; Tester, J.W. Thermochemical biofuel production in hydrothermal media: A review of sub- and supercritical water technologies. *Energy Environ. Sci.* **2008**, *1*, 32–65. [\[CrossRef\]](#)
75. Toor, S.S.; Rosendahl, L.; Rudolf, A. Hydrothermal liquefaction of biomass: A review of subcritical water technologies. *Energy* **2011**, *36*, 2328–2342. [\[CrossRef\]](#)
76. Ruiz, H.A.; Rodríguez-Jasso, R.M.; Fernandes, B.D.; Vicente, A.A.; Teixeira, J.A. Hydrothermal processing, as an alternative for upgrading agriculture residues and marine biomass according to the biorefinery concept: A review. *Renew. Sustain. Energy Rev.* **2013**, *21*, 35–51. [\[CrossRef\]](#)
77. Knez, Ž.; Markočič, E.; Hrnčič, M.K.; Ravber, M.; Škerget, M. High pressure water reforming of biomass for energy and chemicals: A short review. *J. Supercrit. Fluids* **2015**, *96*, 46–52. [\[CrossRef\]](#)
78. Pan, H. Synthesis of polymers from organic solvent liquefied biomass: A review. *Renew. Sustain. Energy Rev.* **2011**, *15*, 3454–3463. [\[CrossRef\]](#)
79. Dimitriadis, A.; Bezergianni, S. Hydrothermal liquefaction of various biomass and waste feedstocks for biocrude production: A state of the art review. *Renew. Sustain. Energy Rev.* **2017**, *68*, 113–125. [\[CrossRef\]](#)
80. Kim, J.-Y.; Lee, H.W.; Lee, S.M.; Jae, J.; Park, Y.-K. Overview of the recent advances in lignocellulose liquefaction for producing biofuels, bio-based materials and chemicals. *Bioresour. Technol.* **2019**, *279*, 373–384. [\[CrossRef\]](#)
81. Haverly, M.R.; Schulz, T.C.; Whitmer, L.E.; Friend, A.J.; Funkhouser, J.M.; Smith, R.G.; Young, M.K.; Brown, R.C. Continuous solvent liquefaction of biomass in a hydrocarbon solvent. *Fuel* **2018**, *211*, 291–300. [\[CrossRef\]](#)
82. Tran, K.-Q. Fast hydrothermal liquefaction for production of chemicals and biofuels from wet biomass—The need to develop a plug-flow reactor. *Bioresour. Technol.* **2016**, *213*, 327–332. [\[CrossRef\]](#) [\[PubMed\]](#)
83. Sinan, N.; Unur, E. Hydrothermal conversion of lignocellulosic biomass into high-value energy storage materials. *J. Energy Chem.* **2017**, *26*, 783–789. [\[CrossRef\]](#)
84. Ravichandran, S.R.; Venkatachalam, C.D.; Sengottian, M.; Sekar, S.; Kandasamy, S.; Ramasamy Subramanian, K.P.; Purushothaman, K.; Lavanya Chandrasekaran, A.; Narayanan, M. A review on hydrothermal liquefaction of algal biomass on process parameters, purification and applications. *Fuel* **2022**, *313*, 122679. [\[CrossRef\]](#)
85. Zhu, Z.; Rosendahl, L.; Toor, S.S.; Yu, D.; Chen, G. Hydrothermal liquefaction of barley straw to bio-crude oil: Effects of reaction temperature and aqueous phase recirculation. *Appl. Energy* **2015**, *137*, 183–192. [\[CrossRef\]](#)
86. El Bast, M.; Allam, N.; Abou Msallem, Y.; Awad, S.; Loubar, K. A review on continuous biomass hydrothermal liquefaction systems: Process design and operating parameters effects on biocrude. *J. Energy Inst.* **2023**, *108*, 101260. [\[CrossRef\]](#)

87. Hao, B.; Xu, D.; Wei, Y.; Diao, Y.; Yang, L.; Fan, L.; Guo, Y. Mathematical models application in optimization of hydrothermal liquefaction of biomass. *Fuel Process. Technol.* **2023**, *243*, 107673. [\[CrossRef\]](#)
88. Li, L.; Dong, S.; Li, H.; Wang, M.; Liang, L.; Pang, M. Introducing hydroxyl-containing POSS to liquefaction of biomass: An alternative route to bio-based polyurethane nanocomposites for coated fertilizers. *Eur. Polym. J.* **2023**, *192*, 112082. [\[CrossRef\]](#)
89. Chen, Z.; Long, J. Organosolv liquefaction of sugarcane bagasse catalyzed by acidic ionic liquids. *Bioresour. Technol.* **2016**, *214*, 16–23. [\[CrossRef\]](#)
90. Hafez, I.; Hassan, E.B. Rapid liquefaction of giant miscanthus feedstock in ethanol–water system for production of biofuels. *Energy Convers. Manag.* **2015**, *91*, 219–224. [\[CrossRef\]](#)
91. Grilc, M.; Likozar, B.; Levec, J. Simultaneous Liquefaction and Hydrodeoxygenation of Lignocellulosic Biomass over NiMo/Al₂O₃, Pd/Al₂O₃, and Zeolite Y Catalysts in Hydrogen Donor Solvents. *ChemCatChem* **2015**, *8*, 180–191. [\[CrossRef\]](#)
92. Govindasamy, G.; Sharma, R.; Subramanian, S. Studies on the effect of heterogeneous catalysts on the hydrothermal liquefaction of sugarcane bagasse to low-oxygen-containing bio-oil. *Biofuels* **2018**, *10*, 665–675. [\[CrossRef\]](#)
93. Yan, H.-L.; Zong, Z.-M.; Li, Z.-K.; Kong, J.; Zheng, Q.-X.; Li, Y.; Wei, X.-Y. Sweet sorghum stalk liquefaction in supercritical methanol: Effects of operating conditions on product yields and molecular composition of soluble fraction. *Fuel Process. Technol.* **2017**, *155*, 42–50. [\[CrossRef\]](#)
94. Oliveira, A.L.P.C.; Almeida, P.S.; Campos, M.C.V.; Franceschi, E.; Dariva, C.; Borges, G.R. Thermoliquefaction of palm oil fiber (*Elaeis* sp.) using supercritical ethanol. *Bioresour. Technol.* **2017**, *230*, 1–7. [\[CrossRef\]](#) [\[PubMed\]](#)
95. Yue, Y.; Kastner, J.R.; Mani, S. Two-Stage Hydrothermal Liquefaction of Sweet Sorghum Biomass—Part 1: Production of Sugar Mixtures. *Energy Fuels* **2018**, *32*, 7611–7619. [\[CrossRef\]](#)
96. Pan, Z.-Q.; Huang, H.-J.; Zhou, C.-F.; Xiao, X.-F.; He, X.-W.; Lai, F.-Y.; Xiong, J.-B. Highly efficient conversion of camphor tree sawdust into bio-oil and biochar products by liquefaction in ethanol-water cosolvent. *J. Anal. Appl. Pyrolysis* **2018**, *136*, 186–198. [\[CrossRef\]](#)
97. Nguyen, T.A.; Ha, T.M.N.; Nguyen, B.T.; Ha, D.; Vu Vo, T.; Nguyen, D.M.; Vo, D.K.; Nguyen, N.T.; Nguyen, T.V.; Hoang, D. Microwave-assisted polyol liquefaction from bamboo for bio-polyurethane foams fabrication. *J. Environ. Chem. Eng.* **2023**, *11*, 109605. [\[CrossRef\]](#)
98. Feng, S.; Wei, R.; Leitch, M.; Xu, C.C. Comparative study on lignocellulose liquefaction in water, ethanol, and water/ethanol mixture: Roles of ethanol and water. *Energy* **2018**, *155*, 234–241. [\[CrossRef\]](#)
99. Zhou, R.; Zhou, R.; Wang, S.; Lan, Z.; Zhang, X.; Yin, Y.; Tu, S.; Yang, S.; Ye, L. Fast liquefaction of bamboo shoot shell with liquid-phase microplasma assisted technology. *Bioresour. Technol.* **2016**, *218*, 1275–1278. [\[CrossRef\]](#)
100. Chen, K.; Lyu, H.; Hao, S.; Luo, G.; Zhang, S.; Chen, J. Separation of phenolic compounds with modified adsorption resin from aqueous phase products of hydrothermal liquefaction of rice straw. *Bioresour. Technol.* **2015**, *182*, 160–168. [\[CrossRef\]](#)
101. Hwang, H.; Lee, J.H.; Choi, I.-G.; Choi, J.W. Comprehensive characterization of hydrothermal liquefaction products obtained from woody biomass under various alkali catalyst concentrations. *Environ. Technol.* **2018**, *40*, 1657–1667. [\[CrossRef\]](#) [\[PubMed\]](#)
102. Pedersen, T.H.; Grigoros, I.F.; Hoffmann, J.; Toor, S.S.; Daraban, I.M.; Jensen, C.U.; Iversen, S.B.; Madsen, R.B.; Glasius, M.; Arturi, K.R.; et al. Continuous hydrothermal co-liquefaction of aspen wood and glycerol with water phase recirculation. *Appl. Energy* **2016**, *162*, 1034–1041. [\[CrossRef\]](#)
103. Wu, X.-F.; Zhou, Q.; Li, M.-F.; Li, S.-X.; Bian, J.; Peng, F. Conversion of poplar into bio-oil via subcritical hydrothermal liquefaction: Structure and antioxidant capacity. *Bioresour. Technol.* **2018**, *270*, 216–222. [\[CrossRef\]](#) [\[PubMed\]](#)
104. Liu, H.-M.; Li, H.-Y.; Li, M.-F. Cornstalk liquefaction in sub- and super-critical ethanol: Characterization of solid residue and the liquefaction mechanism. *J. Energy Inst.* **2017**, *90*, 734–742. [\[CrossRef\]](#)
105. Akalin, M.K.; Das, P.; Alper, K.; Tekin, K.; Ragauskas, A.J.; Karagöz, S. Deconstruction of lignocellulosic biomass with hydrated cerium (III) chloride in water and ethanol. *Appl. Catal. A Gen.* **2017**, *546*, 67–78. [\[CrossRef\]](#)
106. Xu, J.; Xie, X.; Wang, J.; Jiang, J. Directional liquefaction coupling fractionation of lignocellulosic biomass for platform chemicals. *Green Chem.* **2016**, *18*, 3124–3138. [\[CrossRef\]](#)
107. Tekin, K.; Akalin, M.K.; Karagöz, S. The effects of water tolerant Lewis acids on the hydrothermal liquefaction of lignocellulosic biomass. *J. Energy Inst.* **2016**, *89*, 627–635. [\[CrossRef\]](#)
108. Li, Q.; Liu, D.; Song, L.; Hou, X.; Wu, C.; Yan, Z. Efficient hydro-liquefaction of woody biomass over ionic liquid nickel based catalyst. *Ind. Crops Prod.* **2018**, *113*, 157–166. [\[CrossRef\]](#)
109. Lu, Z.; Fan, L.; Wu, Z.; Zhang, H.; Liao, Y.; Zheng, D.; Wang, S. Efficient liquefaction of woody biomass in polyhydric alcohol with acidic ionic liquid as a green catalyst. *Biomass Bioenergy* **2015**, *81*, 154–161. [\[CrossRef\]](#)
110. Singh, R.; Prakash, A.; Balagurumurthy, B.; Singh, R.; Saran, S.; Bhaskar, T. Hydrothermal liquefaction of agricultural and forest biomass residue: Comparative study. *J. Mater. Cycles Waste Manag.* **2014**, *17*, 442–452. [\[CrossRef\]](#)
111. Singh, R.; Chaudhary, K.; Biswas, B.; Balagurumurthy, B.; Bhaskar, T. Hydrothermal liquefaction of rice straw: Effect of reaction environment. *J. Supercrit. Fluids* **2015**, *104*, 70–75. [\[CrossRef\]](#)
112. Nazari, L.; Yuan, Z.; Souzanchi, S.; Ray, M.B.; Xu, C. Hydrothermal liquefaction of woody biomass in hot-compressed water: Catalyst screening and comprehensive characterization of bio-crude oils. *Fuel* **2015**, *162*, 74–83. [\[CrossRef\]](#)
113. Zhao, T.; Zhang, Y.; Zhao, G.; Chen, X.; Han, L.; Xiao, W. Impact of biomass feedstock variability on acid-catalyzed alcoholysis performance. *Fuel Process. Technol.* **2018**, *180*, 14–22. [\[CrossRef\]](#)

114. Jensen, C.U.; Rosendahl, L.A.; Olofsson, G. Impact of nitrogenous alkaline agent on continuous HTL of lignocellulosic biomass and biocrude upgrading. *Fuel Process. Technol.* **2017**, *159*, 376–385. [\[CrossRef\]](#)
115. Deng, H.; Meredith, W.; Uguna, C.N.; Snape, C.E. Impact of solvent type and condition on biomass liquefaction to produce heavy oils in high yield with low oxygen contents. *J. Anal. Appl. Pyrolysis* **2015**, *113*, 340–348. [\[CrossRef\]](#)
116. Hu, Y.; Feng, S.; Bassi, A.; Xu, C. Improvement in bio-crude yield and quality through co-liquefaction of algal biomass and sawdust in ethanol-water mixed solvent and recycling of the aqueous by-product as a reaction medium. *Energy Convers. Manag.* **2018**, *171*, 618–625. [\[CrossRef\]](#)
117. Zhu, Z.; Toor, S.S.; Rosendahl, L.; Yu, D.; Chen, G. Influence of alkali catalyst on product yield and properties via hydrothermal liquefaction of barley straw. *Energy* **2015**, *80*, 284–292. [\[CrossRef\]](#)
118. Brand, S.; Kim, J. Liquefaction of major lignocellulosic biomass constituents in supercritical ethanol. *Energy* **2015**, *80*, 64–74. [\[CrossRef\]](#)
119. Koriakin, A.; Moon, S.; Kim, D.-W.; Lee, C.-H. Liquefaction of oil palm empty fruit bunch using sub- and supercritical tetralin, n-dodecane, and their mixture. *Fuel* **2017**, *208*, 184–192. [\[CrossRef\]](#)
120. Zhai, Q.; Li, F.; Wang, F.; Xu, J.; Jiang, J.; Cai, Z. Liquefaction of poplar biomass for value-added platform chemicals. *Cellulose* **2018**, *25*, 4663–4675. [\[CrossRef\]](#)
121. Miyata, Y.; Sagata, K.; Yamazaki, Y.; Teramura, H.; Hirano, Y.; Ogino, C.; Kita, Y. Mechanism of the Fe-Assisted Hydrothermal Liquefaction of Lignocellulosic Biomass. *Ind. Eng. Chem. Res.* **2018**, *57*, 14870–14877. [\[CrossRef\]](#)
122. Yim, S.C.; Quitain, A.T.; Yusup, S.; Sasaki, M.; Uemura, Y.; Kida, T. Metal oxide-catalyzed hydrothermal liquefaction of Malaysian oil palm biomass to bio-oil under supercritical condition. *J. Supercrit. Fluids* **2017**, *120*, 384–394. [\[CrossRef\]](#)
123. Miličević, A.; Belošević, S.; Žarković, M.; Tomanović, I.; Crnomarković, N.; Stojanović, A.; Stupar, G.; Deng, L.; Che, D. Effects of biomass particles size and shape on combustion process in the swirl-stabilized burner reactor: CFD and machine learning approach. *Biomass Bioenergy* **2023**, *174*, 106817. [\[CrossRef\]](#)
124. Drobnjak, A.; Jelonek, Z.; Mastalerz, M.; Jelonek, I. Residential gasification of solid biomass: Influence of raw material on emissions. *Int. J. Coal Geol.* **2023**, *271*, 104247. [\[CrossRef\]](#)
125. Faizan, M.; Song, H. Critical review on catalytic biomass gasification: State-of-Art progress, technical challenges, and perspectives in future development. *J. Clean. Prod.* **2023**, *408*, 137224. [\[CrossRef\]](#)
126. Tezer, Ö.; Karabağ, N.; Öngen, A.; Çolpan, C.Ö.; Ayol, A. Biomass gasification for sustainable energy production: A review. *Int. J. Hydrogen Energy* **2022**, *47*, 15419–15433. [\[CrossRef\]](#)
127. Yu, H.; Wu, Z.; Chen, G. Catalytic gasification characteristics of cellulose, hemicellulose and lignin. *Renew. Energy* **2018**, *121*, 559–567. [\[CrossRef\]](#)
128. Demirbas, A. Comparison of thermochemical conversion processes of biomass to hydrogen-rich gas mixtures. *Energy Sources Part A Recovery Util. Environ. Eff.* **2016**, *38*, 2971–2976. [\[CrossRef\]](#)
129. Cao, L.; Yu, I.K.M.; Xiong, X.; Tsang, D.C.W.; Zhang, S.; Clark, J.H.; Hu, C.; Ng, Y.H.; Shang, J.; Ok, Y.S. Biorenewable hydrogen production through biomass gasification: A review and future prospects. *Environ. Res.* **2020**, *186*, 109547. [\[CrossRef\]](#)
130. Verma, S.; Dregulo, A.M.; Kumar, V.; Bhargava, P.C.; Khan, N.; Singh, A.; Sun, X.; Sindhu, R.; Binod, P.; Zhang, Z.; et al. Reaction engineering during biomass gasification and conversion to energy. *Energy* **2023**, *266*, 126458. [\[CrossRef\]](#)
131. Fremaux, S.; Beheshti, S.-M.; Ghassemi, H.; Shahsavani-Markadeh, R. An experimental study on hydrogen-rich gas production via steam gasification of biomass in a research-scale fluidized bed. *Energy Convers. Manag.* **2015**, *91*, 427–432. [\[CrossRef\]](#)
132. Berdugo Vilches, T.; Marinkovic, J.; Seemann, M.; Thunman, H. Comparing Active Bed Materials in a Dual Fluidized Bed Biomass Gasifier: Olivine, Bauxite, Quartz-Sand, and Ilmenite. *Energy Fuels* **2016**, *30*, 4848–4857. [\[CrossRef\]](#)
133. Rapagnà, S.; Gallucci, K.; Foscolo, P.U. Olivine, dolomite and ceramic filters in one vessel to produce clean gas from biomass. *Waste Manag.* **2018**, *71*, 792–800. [\[CrossRef\]](#) [\[PubMed\]](#)
134. Hernández, J.J.; Aranda, G.; Barba, J.; Mendoza, J.M. Effect of steam content in the air–steam flow on biomass entrained flow gasification. *Fuel Process. Technol.* **2012**, *99*, 43–55. [\[CrossRef\]](#)
135. Iovane, P.; Donatelli, A.; Molino, A. Influence of feeding ratio on steam gasification of palm shells in a rotary kiln pilot plant. Experimental and numerical investigations. *Biomass Bioenergy* **2013**, *56*, 423–431. [\[CrossRef\]](#)
136. Schweitzer, D.; Gredinger, A.; Schmid, M.; Waizmann, G.; Beiror, M.; Spörl, R.; Scheffknecht, G. Steam gasification of wood pellets, sewage sludge and manure: Gasification performance and concentration of impurities. *Biomass Bioenergy* **2018**, *111*, 308–319. [\[CrossRef\]](#)
137. Niu, Y.; Han, F.; Chen, Y.; Lyu, Y.; Wang, L. Experimental study on steam gasification of pine particles for hydrogen-rich gas. *J. Energy Inst.* **2017**, *90*, 715–724. [\[CrossRef\]](#)
138. Zhang, Z.; Pang, S. Experimental investigation of biomass devolatilization in steam gasification in a dual fluidised bed gasifier. *Fuel* **2017**, *188*, 628–635. [\[CrossRef\]](#)
139. Erkiaga, A.; Lopez, G.; Amutio, M.; Bilbao, J.; Olazar, M. Influence of operating conditions on the steam gasification of biomass in a conical spouted bed reactor. *Chem. Eng. J.* **2014**, *237*, 259–267. [\[CrossRef\]](#)
140. Erkiaga, A.; Lopez, G.; Amutio, M.; Bilbao, J.; Olazar, M. Steam gasification of biomass in a conical spouted bed reactor with olivine and γ -alumina as primary catalysts. *Fuel Process. Technol.* **2013**, *116*, 292–299. [\[CrossRef\]](#)
141. Erkiaga, A.; Lopez, G.; Amutio, M.; Bilbao, J.; Olazar, M. Syngas from steam gasification of polyethylene in a conical spouted bed reactor. *Fuel* **2013**, *109*, 461–469. [\[CrossRef\]](#)

142. Sansaniwal, S.K.; Pal, K.; Rosen, M.A.; Tyagi, S.K. Recent advances in the development of biomass gasification technology: A comprehensive review. *Renew. Sustain. Energy Rev.* **2017**, *72*, 363–384. [\[CrossRef\]](#)
143. Puig-Arnavat, M.; Bruno, J.C.; Coronas, A. Review and analysis of biomass gasification models. *Renew. Sustain. Energy Rev.* **2010**, *14*, 2841–2851. [\[CrossRef\]](#)
144. Valizadeh, S.; Hakimian, H.; Farooq, A.; Jeon, B.-H.; Chen, W.-H.; Hoon Lee, S.; Jung, S.-C.; Won Seo, M.; Park, Y.-K. Valorization of biomass through gasification for green hydrogen generation: A comprehensive review. *Bioresour. Technol.* **2022**, *365*, 128143. [\[CrossRef\]](#) [\[PubMed\]](#)
145. Xia, Y.; Zhang, J.; Tang, C.; Pan, W. Research and application of online monitoring of coal and biomass co-combustion and biomass combustion characteristics based on combustion flame. *J. Energy Inst.* **2023**, *108*, 101191. [\[CrossRef\]](#)
146. Zha, Z.; Ge, Z.; Wang, S.; Li, F.; Ma, Y.; Wu, Y.; Zeng, M.; Huang, C.; Zhang, H. Biomass combustion behaviors based on online monitoring system: Morphological, thermodynamic and kinetic analysis. *Combust. Flame* **2023**, *252*, 112750. [\[CrossRef\]](#)
147. Gorbounov, M.; Petrovic, B.; Ozmen, S.; Clough, P.; Masoudi Soltani, S. Activated carbon derived from Biomass combustion bottom ash as solid sorbent for CO₂ adsorption. *Chem. Eng. Res. Des.* **2023**, *194*, 325–343. [\[CrossRef\]](#)
148. Kopczyński, M.; Lasek, J.A.; Iluk, A.; Zuwała, J. The co-combustion of hard coal with raw and torrefied biomasses (willow (*Salix viminalis*), olive oil residue and waste wood from furniture manufacturing). *Energy* **2017**, *140*, 1316–1325. [\[CrossRef\]](#)
149. Abuelnuor, A.A.A.; Wahid, M.A.; Hosseini, S.E.; Saat, A.; Saqr, K.M.; Sait, H.H.; Osman, M. Characteristics of biomass in flameless combustion: A review. *Renew. Sustain. Energy Rev.* **2014**, *33*, 363–370. [\[CrossRef\]](#)
150. Mian, I.; Li, X.; Dacres, O.D.; Wang, J.; Wei, B.; Jian, Y.; Zhong, M.; Liu, J.; Ma, F.; Rahman, N. Combustion kinetics and mechanism of biomass pellet. *Energy* **2020**, *205*, 117909. [\[CrossRef\]](#)
151. Magdziarz, A.; Wilk, M. Thermal characteristics of the combustion process of biomass and sewage sludge. *J. Therm. Anal. Calorim.* **2013**, *114*, 519–529. [\[CrossRef\]](#)
152. Li, J.; Paul, M.C.; Younger, P.L.; Watson, I.; Hossain, M.; Welch, S. Characterization of biomass combustion at high temperatures based on an upgraded single particle model. *Appl. Energy* **2015**, *156*, 749–755. [\[CrossRef\]](#)
153. Toptas, A.; Yildirim, Y.; Duman, G.; Yanik, J. Combustion behavior of different kinds of torrefied biomass and their blends with lignite. *Bioresour. Technol.* **2015**, *177*, 328–336. [\[CrossRef\]](#) [\[PubMed\]](#)
154. Odziejewicz, J.I.; Wołejko, E.; Wydro, U.; Wasił, M.; Jabłońska-Trypuć, A. Utilization of Ashes from Biomass Combustion. *Energies* **2022**, *15*, 9653. [\[CrossRef\]](#)
155. Vasilatos, C.; Kypridou, Z.; Anastasatou, M.; Aspiotis, K. Sustainable Restoration of Depleted Quarries by the Utilization of Biomass Energy By-Products: The Case of Olive Kernel Residuals. *Sustainability* **2023**, *15*, 1642. [\[CrossRef\]](#)
156. Makarfi Isa, Y.; Ganda, E.T. Bio-oil as a potential source of petroleum range fuels. *Renew. Sustain. Energy Rev.* **2018**, *81*, 69–75. [\[CrossRef\]](#)
157. Hu, X.; Gholizadeh, M. Progress of the applications of bio-oil. *Renew. Sustain. Energy Rev.* **2020**, *134*, 110124. [\[CrossRef\]](#)
158. Mutsengerere, S.; Chihobo, C.H.; Musademba, D.; Nhapi, I. A review of operating parameters affecting bio-oil yield in microwave pyrolysis of lignocellulosic biomass. *Renew. Sustain. Energy Rev.* **2019**, *104*, 328–336. [\[CrossRef\]](#)
159. Ferreira-Santos, P.; Zanusso, E.; Genisheva, Z.; Rocha, C.M.R.; Teixeira, J.A. Green and Sustainable Valorization of Bioactive Phenolic Compounds from Pinus By-Products. *Molecules* **2020**, *25*, 2931. [\[CrossRef\]](#)
160. Ren, J.; Liu, Y.-L. Progress and prospects of produced gas utilization from biomass tar reforming. *J. Hazard. Mater. Lett.* **2020**, *1*, 100008. [\[CrossRef\]](#)
161. Yan, B.; Jia, X.; Li, J.; Li, Z.; Che, Y.; Zhou, Z.; Zhao, J.; Zhou, S.; Chen, G. In-situ elimination of biomass gasification tar based on the understanding of tar formation process: A review. *J. Energy Inst.* **2024**, *112*, 101477. [\[CrossRef\]](#)
162. Bhaskaran, A.; Sharma, D.; Roy, S.; Singh, S.A. Technological solutions for NO_x, SO_x, and VOC abatement: Recent breakthroughs and future directions. *Environ. Sci. Pollut. Res.* **2023**, *30*, 91501–91533. [\[CrossRef\]](#) [\[PubMed\]](#)
163. Tomlin, A.S. Air Quality and Climate Impacts of Biomass Use as an Energy Source: A Review. *Energy Fuels* **2021**, *35*, 14213–14240. [\[CrossRef\]](#)
164. Yao, X.; Liu, Q.; Kang, Z.; An, Z.; Zhou, H.; Xu, K. Quantitative study on thermal conversion behaviours and gas emission properties of biomass in nitrogen and in CO₂/N₂ mixtures by TGA/DTG and a fixed-bed tube furnace. *Energy* **2023**, *270*, 126904. [\[CrossRef\]](#)
165. Mao, H.; Jiang, H.; Sang, L.; Li, S.; Chen, F.-X.; Zhao, Z.-P. An integrated ionic liquid hybrid polymeric fixed-bed reactor with a porous sponge-like structure for biomass hydrolysis. *Chem. Eng. Sci.* **2022**, *253*, 117562. [\[CrossRef\]](#)
166. Fernandez, E.; Santamaria, L.; García, I.; Amutio, M.; Artetxe, M.; Lopez, G.; Bilbao, J.; Olazar, M. Elucidating coke formation and evolution in the catalytic steam reforming of biomass pyrolysis volatiles at different fixed bed locations. *Chin. J. Catal.* **2023**, *48*, 101–116. [\[CrossRef\]](#)
167. Ku, X.; Shen, F.; Jin, H.; Lin, J.; Li, H. Simulation of Biomass Pyrolysis in a Fluidized Bed Reactor Using Thermally Thick Treatment. *Ind. Eng. Chem. Res.* **2019**, *58*, 1720–1731. [\[CrossRef\]](#)
168. Jeong, J.-Y.; Yang, C.-W.; Lee, U.-D.; Jeong, S.-H. Characteristics of the pyrolytic products from the fast pyrolysis of palm kernel cake in a bench-scale fluidized bed reactor. *J. Anal. Appl. Pyrolysis* **2020**, *145*, 104708. [\[CrossRef\]](#)
169. Kraft, S.; Kirnbauer, F.; Hofbauer, H. CPFD simulations of an industrial-sized dual fluidized bed steam gasification system of biomass with 8 MW fuel input. *Appl. Energy* **2017**, *190*, 408–420. [\[CrossRef\]](#)

170. Salema, A.A.; Afzal, M.T.; Bennamoun, L. Pyrolysis of corn stalk biomass briquettes in a scaled-up microwave technology. *Bioresour. Technol.* **2017**, *233*, 353–362. [[CrossRef](#)]
171. Omoriyekomwan, J.E.; Tahmasebi, A.; Dou, J.; Wang, R.; Yu, J. A review on the recent advances in the production of carbon nanotubes and carbon nanofibers via microwave-assisted pyrolysis of biomass. *Fuel Process. Technol.* **2021**, *214*, 106686. [[CrossRef](#)]
172. Zhao, X.; Wang, W.; Liu, H.; Ma, C.; Song, Z. Microwave pyrolysis of wheat straw: Product distribution and generation mechanism. *Bioresour. Technol.* **2014**, *158*, 278–285. [[CrossRef](#)] [[PubMed](#)]
173. Shuping, Z.; Yulong, W.; Mingde, Y.; Chun, L.; Junmao, T. Pyrolysis characteristics and kinetics of the marine microalgae *Dunaliella tertiolecta* using thermogravimetric analyzer. *Bioresour. Technol.* **2010**, *101*, 359–365. [[CrossRef](#)] [[PubMed](#)]
174. Demiral, İ.; Şensöz, S. Fixed-Bed Pyrolysis of Hazelnut (*Corylus avellana* L.) Bagasse: Influence of Pyrolysis Parameters on Product Yields. *Energy Sources Part A Recovery Util. Environ. Eff.* **2006**, *28*, 1149–1158.
175. Pütün, A.E.; Özbay, N.; Önal, E.P.; Pütün, E. Fixed-bed pyrolysis of cotton stalk for liquid and solid products. *Fuel Process. Technol.* **2005**, *86*, 1207–1219. [[CrossRef](#)]
176. Xue, A.J.; Pan, J.H.; Tian, M.C. Experimental Study of Impact of Biomass Pellet Size on the Pyrolysis Products. *Adv. Mater. Res.* **2013**, *641–642*, 756–759. [[CrossRef](#)]
177. Shen, J.; Wang, X.-S.; Garcia-Perez, M.; Mourant, D.; Rhodes, M.J.; Li, C.-Z. Effects of particle size on the fast pyrolysis of oil mallee woody biomass. *Fuel* **2009**, *88*, 1810–1817. [[CrossRef](#)]
178. Wang, X.; Kersten, S.R.A.; Prins, W.; van Swaaij, W.P.M. Biomass Pyrolysis in a Fluidized Bed Reactor. Part 2: Experimental Validation of Model Results. *Ind. Eng. Chem. Res.* **2005**, *44*, 8786–8795. [[CrossRef](#)]
179. Erlich, C.; Bjornbom, E.; Bolado, D.; Giner, M.; Fransson, T. Pyrolysis and gasification of pellets from sugar cane bagasse and wood. *Fuel* **2006**, *85*, 1535–1540. [[CrossRef](#)]
180. Karkania, V.; Fanara, E.; Zabaniotou, A. Review of sustainable biomass pellets production—A study for agricultural residues pellets' market in Greece. *Renew. Sustain. Energy Rev.* **2012**, *16*, 1426–1436. [[CrossRef](#)]
181. Isaksson, J.; Åsblad, A.; Berntsson, T. Influence of dryer type on the performance of a biomass gasification combined cycle co-located with an integrated pulp and paper mill. *Biomass Bioenergy* **2013**, *59*, 336–347. [[CrossRef](#)]
182. Garcia-Perez, M.; Wang, X.S.; Shen, J.; Rhodes, M.J.; Tian, F.; Lee, W.-J.; Wu, H.; Li, C.-Z. Fast Pyrolysis of Oil Mallee Woody Biomass: Effect of Temperature on the Yield and Quality of Pyrolysis Products. *Ind. Eng. Chem. Res.* **2008**, *47*, 1846–1854. [[CrossRef](#)]
183. Westerhof, R.J.M.; Brilman, D.W.F.; van Swaaij, W.P.M.; Kersten, S.R.A. Effect of Temperature in Fluidized Bed Fast Pyrolysis of Biomass: Oil Quality Assessment in Test Units. *Ind. Eng. Chem. Res.* **2010**, *49*, 1160–1168. [[CrossRef](#)]
184. Li, J.; Yan, R.; Xiao, B.; Wang, X.; Yang, H. Influence of Temperature on the Formation of Oil from Pyrolyzing Palm Oil Wastes in a Fixed Bed Reactor. *Energy Fuels* **2007**, *21*, 2398–2407. [[CrossRef](#)]
185. Ateş, F.; Işıkdag, M.A. Evaluation of the Role of the Pyrolysis Temperature in Straw Biomass Samples and Characterization of the Oils by GC/MS. *Energy Fuels* **2008**, *22*, 1936–1943. [[CrossRef](#)]
186. Fatehi, H.; Qu, Z.; Schmidt, F.M.; Bai, X.-S. Effect of Volatile Reactions on the Thermochemical Conversion of Biomass Particles. *Energy Procedia* **2017**, *105*, 4648–4654. [[CrossRef](#)]
187. Park, S.-W.; Jang, C.-H. Effects of pyrolysis temperature on changes in fuel characteristics of biomass char. *Energy* **2012**, *39*, 187–195. [[CrossRef](#)]
188. Lewandowski, W.M.; Ryms, M.; Kosakowski, W. Thermal Biomass Conversion: A Review. *Processes* **2020**, *8*, 516. [[CrossRef](#)]
189. Seow, Y.X.; Tan, Y.H.; Mubarak, N.M.; Kannedo, J.; Khalid, M.; Ibrahim, M.L.; Ghasemi, M. A review on biochar production from different biomass wastes by recent carbonization technologies and its sustainable applications. *J. Environ. Chem. Eng.* **2022**, *10*, 107017. [[CrossRef](#)]
190. Salehi, E.; Abedi, J.; Harding, T. Bio-oil from Sawdust: Pyrolysis of Sawdust in a Fixed-Bed System. *Energy Fuels* **2009**, *23*, 3767–3772. [[CrossRef](#)]
191. Ozbay, N.; Pütün, A.E.; Pütün, E. Bio-oil production from rapid pyrolysis of cottonseed cake: Product yields and compositions. *Int. J. Energy Res.* **2006**, *30*, 501–510. [[CrossRef](#)]
192. Brand, S.; Hardi, F.; Kim, J.; Suh, D.J. Effect of heating rate on biomass liquefaction: Differences between subcritical water and supercritical ethanol. *Energy* **2014**, *68*, 420–427. [[CrossRef](#)]
193. Guo, H.; Zhang, H.; Chen, X.; Zhang, L.; Huang, C.; Li, H.; Peng, F.; Huang, Q.; Xiong, L.; Ouyang, X.; et al. Catalytic upgrading of biopolyols derived from liquefaction of wheat straw over a high-performance and stable supported amorphous alloy catalyst. *Energy Convers. Manag.* **2018**, *156*, 130–139. [[CrossRef](#)]
194. Liu, H.; Tang, Y.; Ma, X.; Yue, W. Catalytic pyrolysis of corncob with Ni/CaO catalysts for hydrogen-rich gas: Synthesis modes and catalyst/biomass ratios. *J. Ind. Eng. Chem.* **2023**, *123*, 51–61. [[CrossRef](#)]
195. Singh, M.; Salaudeen, S.A.; Norouzi, O.; Al-Salem, S.M.; Gilroyed, B.H.; Dutta, A. Co-pyrolysis of biomass and tires using commercial zeolite and biochar-based catalyst. *Chem. Eng. Process.-Process Intensif.* **2023**, *187*, 109356. [[CrossRef](#)]
196. Gamliel, D.P.; Du, S.; Bollas, G.M.; Valla, J.A. Investigation of in situ and ex situ catalytic pyrolysis of miscanthus x giganteus using a PyGC-MS microsystem and comparison with a bench-scale spouted-bed reactor. *Bioresour. Technol.* **2015**, *191*, 187–196. [[CrossRef](#)]
197. Xin, X.; Pang, S.; de Miguel Mercader, F.; Torr, K.M. The effect of biomass pretreatment on catalytic pyrolysis products of pine wood by Py-GC/MS and principal component analysis. *J. Anal. Appl. Pyrolysis* **2019**, *138*, 145–153. [[CrossRef](#)]

198. Ong, H.C.; Chen, W.-H.; Farooq, A.; Gan, Y.Y.; Lee, K.T.; Ashokkumar, V. Catalytic thermochemical conversion of biomass for biofuel production: A comprehensive review. *Renew. Sustain. Energy Rev.* **2019**, *113*, 109266. [\[CrossRef\]](#)
199. Cheng, S.; Wei, L.; Julson, J.; Kharel, P.R.; Cao, Y.; Gu, Z. Catalytic liquefaction of pine sawdust for biofuel development on bifunctional Zn/HZSM-5 catalyst in supercritical ethanol. *J. Anal. Appl. Pyrolysis* **2017**, *126*, 257–266. [\[CrossRef\]](#)
200. Cheng, S.; Wei, L.; Rabnawaz, M. Catalytic liquefaction of pine sawdust and in-situ hydrogenation of bio-crude over bifunctional Co-Zn/HZSM-5 catalysts. *Fuel* **2018**, *223*, 252–260. [\[CrossRef\]](#)
201. Escalante, J.; Chen, W.-H.; Tabatabaei, M.; Hoang, A.T.; Kwon, E.E.; Andrew Lin, K.-Y.; Saravanakumar, A. Pyrolysis of lignocellulosic, algal, plastic, and other biomass wastes for biofuel production and circular bioeconomy: A review of thermogravimetric analysis (TGA) approach. *Renew. Sustain. Energy Rev.* **2022**, *169*, 112914. [\[CrossRef\]](#)
202. Brun, B.F.; Nascimento, M.H.C.; Dias, P.A.C.; Marcarini, W.D.; Singh, M.N.; Filgueiras, P.R.; Vassallo, P.F.; Romão, W.; Mill, J.G.; Martin, F.L.; et al. Fast screening using attenuated total reflectance-fourier transform infrared (ATR-FTIR) spectroscopy of patients based on D-dimer threshold value. *Talanta* **2024**, *269*, 125482. [\[CrossRef\]](#) [\[PubMed\]](#)
203. Huang, Y.; Sekyere, D.T.; Zhang, J.; Tian, Y. Fast pyrolysis behaviors of biomass with high contents of ash and nitrogen using TG-FTIR and Py-GC/MS. *J. Anal. Appl. Pyrolysis* **2023**, *170*, 105922. [\[CrossRef\]](#)
204. Anca-Couce, A.; Tsekos, C.; Retschitzegger, S.; Zimbardi, F.; Funke, A.; Banks, S.; Kraia, T.; Marques, P.; Scharler, R.; de Jong, W.; et al. Biomass pyrolysis TGA assessment with an international round robin. *Fuel* **2020**, *276*, 118002. [\[CrossRef\]](#)
205. Lee, C.S.; Sung, T.M.; Kim, H.S.; Jeon, C.H. Classification of forensic soil evidences by application of THM-PyGC/MS and multivariate analysis. *J. Anal. Appl. Pyrolysis* **2012**, *96*, 33–42. [\[CrossRef\]](#)
206. Gao, N.; Li, A.; Quan, C.; Du, L.; Duan, Y. TG-FTIR and Py-GC/MS analysis on pyrolysis and combustion of pine sawdust. *J. Anal. Appl. Pyrolysis* **2013**, *100*, 26–32. [\[CrossRef\]](#)
207. Jiang, X.; Lu, Q.; Hu, B.; Liu, J.; Dong, C.; Yang, Y. Intermolecular interaction mechanism of lignin pyrolysis: A joint theoretical and experimental study. *Fuel* **2018**, *215*, 386–394. [\[CrossRef\]](#)
208. Mahmood, H.; Shakeel, A.; Abdullah, A.; Khan, M.I.; Moniruzzaman, M. A Comparative Study on Suitability of Model-Free and Model-Fitting Kinetic Methods to Non-Isothermal Degradation of Lignocellulosic Materials. *Polymers* **2021**, *13*, 2504. [\[CrossRef\]](#)
209. White, J.E.; Catallo, W.J.; Legendre, B.L. Biomass pyrolysis kinetics: A comparative critical review with relevant agricultural residue case studies. *J. Anal. Appl. Pyrolysis* **2011**, *91*, 1–33. [\[CrossRef\]](#)
210. Luo, H.; Wang, X.; Liu, X.; Wu, X.; Shi, X.; Xiong, Q. A review on CFD simulation of biomass pyrolysis in fluidized bed reactors with emphasis on particle-scale models. *J. Anal. Appl. Pyrolysis* **2022**, *162*, 105433. [\[CrossRef\]](#)
211. Gao, X.; Lu, L.; Shahnam, M.; Rogers, W.A.; Smith, K.; Gaston, K.; Robichaud, D.; Brennan Pecha, M.; Crowley, M.; Ciesielski, P.N.; et al. Assessment of a detailed biomass pyrolysis kinetic scheme in multiscale simulations of a single-particle pyrolyzer and a pilot-scale entrained flow pyrolyzer. *Chem. Eng. J.* **2021**, *418*, 129347. [\[CrossRef\]](#)
212. Liu, H.; Cattolica, R.J.; Seiser, R.; Liao, C.-H. Three-dimensional full-loop simulation of a dual fluidized-bed biomass gasifier. *Appl. Energy* **2015**, *160*, 489–501. [\[CrossRef\]](#)
213. Zhong, W.; Xie, J.; Shao, Y.; Liu, X.; Jin, B. Three-dimensional modeling of olive cake combustion in CFB. *Appl. Therm. Eng.* **2015**, *88*, 322–333. [\[CrossRef\]](#)
214. Oevermann, M.; Gerber, S.; Behrendt, F. Euler–Lagrange/DEM simulation of wood gasification in a bubbling fluidized bed reactor. *Particuology* **2009**, *7*, 307–316. [\[CrossRef\]](#)
215. Loha, C.; Chattopadhyay, H.; Chatterjee, P.K. Three dimensional kinetic modeling of fluidized bed biomass gasification. *Chem. Eng. Sci.* **2014**, *109*, 53–64. [\[CrossRef\]](#)
216. Mahmoudi, A.H.; Hoffmann, F.; Peters, B. Semi-resolved modeling of heat-up, drying and pyrolysis of biomass solid particles as a new feature in XDEM. *Appl. Therm. Eng.* **2016**, *93*, 1091–1104. [\[CrossRef\]](#)
217. Gerber, S.; Oevermann, M. A two dimensional Euler–Lagrangian model of wood gasification in a charcoal bed—Part I: Model description and base scenario. *Fuel* **2014**, *115*, 385–400. [\[CrossRef\]](#)
218. Xie, J.; Zhong, W.; Jin, B.; Shao, Y.; Liu, H. Simulation on gasification of forestry residues in fluidized beds by Eulerian-Lagrangian approach. *Bioresour. Technol.* **2012**, *121*, 36–46. [\[CrossRef\]](#)
219. Yan, L.; Cao, Y.; Zhou, H.; He, B. Investigation on biomass steam gasification in a dual fluidized bed reactor with the granular kinetic theory. *Bioresour. Technol.* **2018**, *269*, 384–392. [\[CrossRef\]](#)
220. Zhong, H.; Xu, F.; Zhang, J.; Zhu, Y.; Liang, S.; Niu, B.; Zhang, X. Variation of Geldart classification in MFM simulation of biomass fast pyrolysis considering the decrease of particle density and diameter. *Renew. Energy* **2019**, *135*, 208–217. [\[CrossRef\]](#)
221. Zhong, H.; Xiong, Q.; Zhu, Y.; Liang, S.; Zhang, J.; Niu, B.; Zhang, X. CFD modeling of the effects of particle shrinkage and intra-particle heat conduction on biomass fast pyrolysis. *Renew. Energy* **2019**, *141*, 236–245. [\[CrossRef\]](#)
222. Han, X.; Liu, Q.; Jiang, X. Heat transfer characteristic of oil shale particle during the retorting. *Int. J. Heat Mass Transf.* **2015**, *84*, 578–583. [\[CrossRef\]](#)
223. Yuan, M.; Chen, N.; Guo, Y.; Chen, B.; Wang, S.; Shen, J.; Qiu, X. Simulation analysis of biomass pyrolysis based on the improved chemical percolation devolatilization model with chain reaction dynamics. *AIChE J.* **2022**, *68*, e17853. [\[CrossRef\]](#)
224. Syamlal, M.; O'Brien, T.J. Simulation of granular layer inversion in liquid fluidized beds. *Int. J. Multiph. Flow* **1988**, *14*, 473–481. [\[CrossRef\]](#)
225. Papadakis, K.; Gu, S.; Bridgwater, A.V. CFD modelling of the fast pyrolysis of biomass in fluidised bed reactors: Modelling the impact of biomass shrinkage. *Chem. Eng. J.* **2009**, *149*, 417–427. [\[CrossRef\]](#)

226. Luo, H.; Lin, W.; Dam-Johansen, K.; Wu, H. Heat-Transfer-Corrected Isothermal Model for Devolatilization of Thermally Thick Biomass Particles. *Energy Fuels* **2020**, *34*, 9620–9631. [\[CrossRef\]](#)
227. Figueroa, J.E.J.; Ardila, Y.C.; Lunelli, B.H.; Filho, R.M.; Wolf Maciel, M.R. Sugarcane bagasse as raw material to Syngas production. In *22nd European Symposium on Computer Aided Process Engineering*; Elsevier: Amsterdam, The Netherlands, 2012; pp. 1118–1122.
228. Xue, Q.; Dalluge, D.; Heindel, T.J.; Fox, R.O.; Brown, R.C. Experimental validation and CFD modeling study of biomass fast pyrolysis in fluidized-bed reactors. *Fuel* **2012**, *97*, 757–769. [\[CrossRef\]](#)
229. Eri, Q.; Wang, B.; Peng, J.; Zhao, X.; Li, T. Detailed CFD modelling of fast pyrolysis of different biomass types in fluidized bed reactors. *Can. J. Chem. Eng.* **2018**, *96*, 2043–2052. [\[CrossRef\]](#)
230. Liu, H.; Cattolica, R.J.; Seiser, R. CFD studies on biomass gasification in a pilot-scale dual fluidized-bed system. *Int. J. Hydrogen Energy* **2016**, *41*, 11974–11989. [\[CrossRef\]](#)
231. Lu, B.; Wang, W.; Li, J. Searching for a mesh-independent sub-grid model for CFD simulation of gas–solid riser flows. *Chem. Eng. Sci.* **2009**, *64*, 3437–3447. [\[CrossRef\]](#)
232. Lu, B.; Wang, W.; Li, J. Eulerian simulation of gas–solid flows with particles of Geldart groups A, B and D using EMMS-based meso-scale model. *Chem. Eng. Sci.* **2011**, *66*, 4624–4635. [\[CrossRef\]](#)
233. Luo, H.; Lu, B.; Zhang, J.; Wu, H.; Wang, W. A grid-independent EMMS/bubbling drag model for bubbling and turbulent fluidization. *Chem. Eng. J.* **2017**, *326*, 47–57. [\[CrossRef\]](#)
234. Wang, W.; Li, J. Simulation of gas–solid two-phase flow by a multi-scale CFD approach—Of the EMMS model to the sub-grid level. *Chem. Eng. Sci.* **2007**, *62*, 208–231. [\[CrossRef\]](#)
235. Xiong, Q.; Kong, S.-C. Modeling effects of interphase transport coefficients on biomass pyrolysis in fluidized beds. *Powder Technol.* **2014**, *262*, 96–105. [\[CrossRef\]](#)
236. Igci, Y.; Andrews, A.T.; Sundaresan, S.; Pannala, S.; O'Brien, T. Filtered two-fluid models for fluidized gas-particle suspensions. *AIChE J.* **2008**, *54*, 1431–1448. [\[CrossRef\]](#)
237. Lu, L.; Gao, X.; Gel, A.; Wiggins, G.M.; Crowley, M.; Pecha, B.; Shahnam, M.; Rogers, W.A.; Parks, J.; Ciesielski, P.N. Investigating biomass composition and size effects on fast pyrolysis using global sensitivity analysis and CFD simulations. *Chem. Eng. J.* **2021**, *421*, 127789. [\[CrossRef\]](#)
238. Lu, B.; Zhang, J.; Luo, H.; Wang, W.; Li, H.; Ye, M.; Liu, Z.; Li, J. Numerical simulation of scale-up effects of methanol-to-olefins fluidized bed reactors. *Chem. Eng. Sci.* **2017**, *171*, 244–255. [\[CrossRef\]](#)
239. Ozarkar, S.S.; Yan, X.; Wang, S.; Milioli, C.C.; Milioli, F.E.; Sundaresan, S. Validation of filtered two-fluid models for gas–particle flows against experimental data from bubbling fluidized bed. *Powder Technol.* **2015**, *284*, 159–169. [\[CrossRef\]](#)
240. Lu, L.; Gao, X.; Dietiker, J.-F.; Shahnam, M.; Rogers, W.A. MFIX based multi-scale CFD simulations of biomass fast pyrolysis: A review. *Chem. Eng. Sci.* **2022**, *248*, 117131. [\[CrossRef\]](#)
241. Luo, H.; Lu, Z.; Jensen, P.A.; Glarborg, P.; Lin, W.; Dam-Johansen, K.; Wu, H. Experimental and modelling study on the influence of wood type, density, water content, and temperature on wood devolatilization. *Fuel* **2020**, *260*, 116410. [\[CrossRef\]](#)
242. Di Blasi, C. Modeling and simulation of combustion processes of charring and non-charring solid fuels. *Prog. Energy Combust. Sci.* **1993**, *19*, 71–104. [\[CrossRef\]](#)
243. Diblasi, C. Modeling chemical and physical processes of wood and biomass pyrolysis. *Prog. Energy Combust. Sci.* **2008**, *34*, 47–90. [\[CrossRef\]](#)
244. Xiong, Q.; Yang, Y.; Xu, F.; Pan, Y.; Zhang, J.; Hong, K.; Lorenzini, G.; Wang, S. Overview of Computational Fluid Dynamics Simulation of Reactor-Scale Biomass Pyrolysis. *ACS Sustain. Chem. Eng.* **2017**, *5*, 2783–2798. [\[CrossRef\]](#)
245. Vandenbroek, R. Biomass combustion for power generation. *Biomass Bioenergy* **1996**, *11*, 271–281. [\[CrossRef\]](#)
246. Xiong, Q.; Zhang, J.; Xu, F.; Wiggins, G.; Stuart Daw, C. Coupling DAEM and CFD for simulating biomass fast pyrolysis in fluidized beds. *J. Anal. Appl. Pyrolysis* **2016**, *117*, 176–181. [\[CrossRef\]](#)
247. Mellin, P.; Kantarelis, E.; Yang, W. Computational fluid dynamics modeling of biomass fast pyrolysis in a fluidized bed reactor, using a comprehensive chemistry scheme. *Fuel* **2014**, *117*, 704–715. [\[CrossRef\]](#)
248. Zhong, H.; Xiong, Q.; Yin, L.; Zhang, J.; Zhu, Y.; Liang, S.; Niu, B.; Zhang, X. CFD-based reduced-order modeling of fluidized-bed biomass fast pyrolysis using artificial neural network. *Renew. Energy* **2020**, *152*, 613–626. [\[CrossRef\]](#)
249. Hooshdaran, B.; Haghshenasfard, M.; Hosseini, S.H.; Esfahany, M.N.; Lopez, G.; Olazar, M. CFD modeling and experimental validation of biomass fast pyrolysis in a conical spouted bed reactor. *J. Anal. Appl. Pyrolysis* **2021**, *154*, 105011. [\[CrossRef\]](#)
250. Sia, S.Q.; Wang, W.-C. Numerical simulations of fluidized bed fast pyrolysis of biomass through computational fluid dynamics. *Renew. Energy* **2020**, *155*, 248–256. [\[CrossRef\]](#)

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.